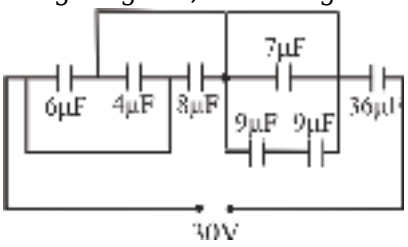
PHYSICS

1) In the following diagram, the charge and potential difference across 8μF capacitance will be



respectively

(1) 80μC, 10 V

(2) 420μC, 20 V

(3) 160μC, 20 V

(4) 360μC, 10 V

2) A body of mass 3kg hits a wall at an angle of 60° with 10 m/sec and returns at same angle with same speed. The impact time was 0.2 sec. Then force exerted on the wall is :-



(1) $150\sqrt{3}\,N$

(2) $50\sqrt{3}\,N$

(3) 100 N

(4) $75\sqrt{3}\,N$

3) A parallel plate condenser has a uniform electric field E(V/m) in the space between the plates. If the distance between the plates is d(m) and area of each plate is A(m^2) the energy (joules) stored in the condenser is

(1) E^2Ad/ϵ_0

(2) $\frac{1}{2}\epsilon_0E^2$

(3) $\epsilon_0E\ Ad$

(4) $\frac{1}{2}\epsilon_0E^2\ Ad$

4) Given the vectors $\vec{a} = -3\hat{i} + 2\hat{j} - \hat{k}$ and $\vec{b} = \hat{i} - 3\hat{j} + 5\hat{k}$, which form two sides of a triangle, what is the area of the triangle?

(1) $7\sqrt{6}$

(2) $\frac{7\sqrt{6}}{2}$

(3) $\sqrt{14}$

(4) $\frac{\sqrt{294}}{4}$

5) A student uses a simple pendulum of exactly 1 m length to determine g, the acceleration due to gravity. He uses a stop watch with the least count of 1 sec for this and record 40 seconds for 20 oscillations for this observation, which of following is true ? (T is time period of oscillations)

$\left[T = 2\pi\sqrt{\frac{L}{g}}\right]$

(1) $\Delta T = 1$ sec ; $\frac{\Delta g}{g} \times 100 = 2.5$ %

(2) $\Delta T = 0.05$ sec ; $\frac{\Delta g}{g} \times 100 = 5$ %

(3) $\Delta T = 0.025$ sec ; $\frac{\Delta g}{g} \times 100 = 5$ %

(4) $\Delta T = 0.025$ sec ; $\frac{\Delta g}{g} \times 100 = 2.5$ %

6) If $K = \frac{d^2x}{dt^2}$ find dimensional formula of K :-

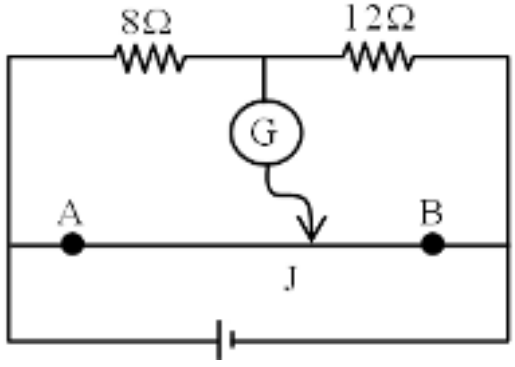
(1) $[L^1 T^{-1}]$

(2) $[L^1 T^{-2}]$

(3) $[L^2 T^{-2}]$

(4) $[L^2 T^{-1}]$

7) The given circuit shows a uniform straight wire AB of 60 cm length fixed at both ends. In order to get zero reading in the galvanometer G, the free end of J is to be placed from B at :



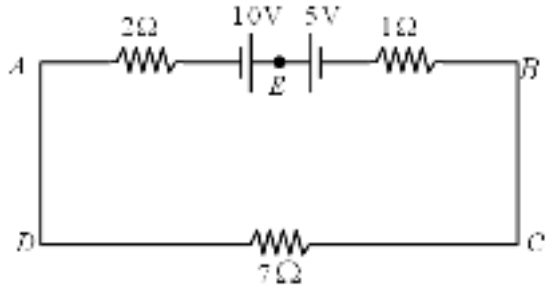
(1) 36 cm

(2) 8 cm

(3) 16 cm

(4) 24 cm

8) Which of the following option is NOT correct: :-



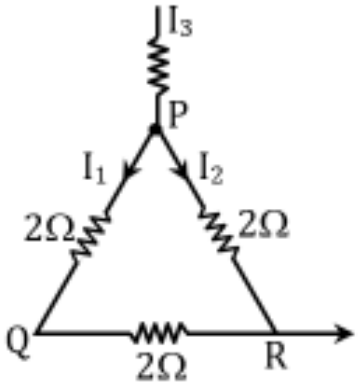
(1) Current is 0.5 A from *A* to *B* through *E*

(2) $V_A - V_E = 9V$

(3) $V_E - V_B = 5.5V$

(4) 5W power is being supplied by 10V source

9) A current I_1 flowing through side PQ of an equilateral triangle PQR having three wires of resistances 2Ω each is 2A. Then the current I_2 and I_3 respectively are :



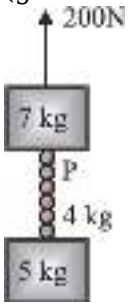
(1) 2A,4A

(2) 4A,6A

(3) 1A,2A

(4) 2A,6A

10) Two blocks of 7 kg and 5 kg are connected by a heavy rope of mass 4 kg. An upward force of 200 N is applied as shown in the diagram. The tension at the top of heavy rope at point P is :-
(g = 10 m/s^2)



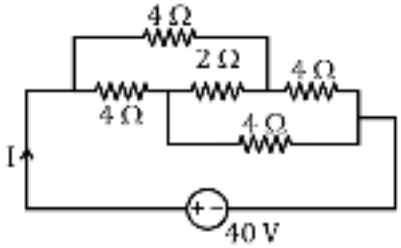
(1) 2.27 N

(2) 112.5 N

(3) 87.5 N

(4) 360 N

11) The current I in the given circuit will be:



(1) 10 A

(2) 20 A

(3) 4 A

(4) 40 A

12) Match the column

Column - I	Column - II
(a) Kirchhoff 1st law	(p) Dependent of physical dimension
(b) Kirchhoff 2nd law	(q) Conservation of charge
(c) Resistance of metal	(r) Conservation of energy
(d) Resistivity of metal	(s) Depends on temperature

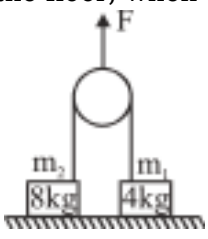
(1) a-q, b-r, c-p,s, d-s

(2) a-p, b-r,s c-q, d-s

(3) a-q, b-r, c-p, d-s

(4) a-q, b-s, c-p, d-r

13) Two block of mass 8 kg and 4 kg are connected by a string as shown in the figure below. Calculate their acceleration, if they are initially at rest on the floor, when a force of 100 N is applied on the massless pulley in upward direction (g = 10 ms^{-2}):-



(1) 0 m/s^2, 0 m/s^2

(2) 0 m/s^2, 2.5 m/s^2

(3) 1.25 m/s^2,2.5 m/s^2

(4) 4.5 m/s^2, 0 m/s^2

14) A capacitor of capacitance C_1 = 1μF can withstand maximum voltage V_1 = 6 kV (kilo-volt) and

another capacitor of capacitance C_2 = 3μF can withstand maximum voltage V_2 = 4kV. When the two capacitors are connected in series, the combined system can withstand a maximum voltage of :-

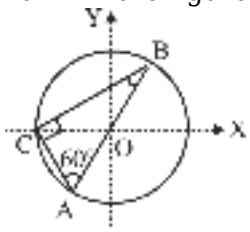
(1) 4 kV

(2) 6 kV

(3) 8 kV

(4) 10 kV

15) Consider a system of three charges $\frac{q}{3}, \frac{q}{3}$ and $-\frac{2q}{3}$ placed at points A, B and C respectively , as shown in the figure. Take O to be the centre of the circle of radius R and angle CAB = 60°. :-



(1) The electric field at point O is $\frac{q}{8\pi\varepsilon_0 R^2}$ directed along the negative x-axis.

(2) The potential energy of the system is zero.

(3) The magnitude of the force between the charges at C and B is $\frac{q^2}{54\pi\varepsilon_0 R^2}$

(4) The potential at point O is $\frac{q}{12\pi\varepsilon_0 R}$

16) The position of a particle moving in the xy-plane at any time t is given by x = ($3t^2$ – 6t) metres, y = (t^2 – 2t) metres. Select the correct statement about the moving particle from the following :-

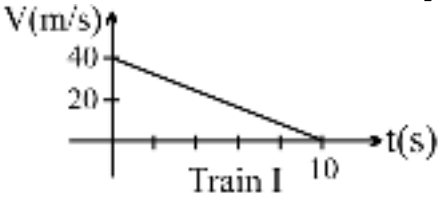
(1) The acceleration of the particle is zero at t = 0 second

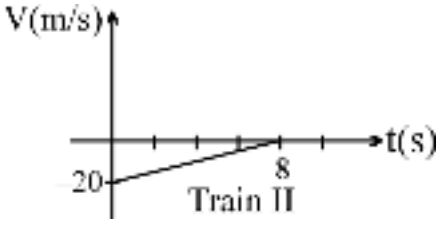
(2) The velocity of the particle is zero at t = 0 second

(3) The velocity of the particle is zero at t = 1 second

(4) The velocity and acceleration of the particle are never zero

17) Two trains, which are moving along different tracks in opposite directions towards each other, are put on the same track by mistake. Their drivers, on noticing the mistake, start slowing down the trains when the trains are 300 m apart. Graphs given below show their velocities as function of time as the trains slow down. The separation between the trains after both have stopped, is :





(1) 120 m

(2) 280 m

(3) 60 m

(4) 20 m

18) If $\vec{A} = \vec{B} + \vec{C}$ and the magnitudes of $\vec{A}, \vec{B}$ and $\vec{C}$ are 5, 4 and 3 units, then the angle between $\vec{A}$ and $\vec{C}$ is :

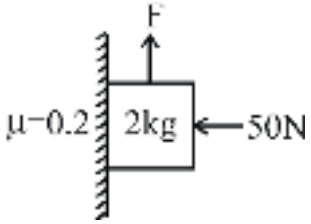
(1) $\cos^{-1}\left(\frac{3}{5}\right)$

(2) $\cos^{-1}\left(\frac{4}{5}\right)$

(3) $\frac{\pi}{2}$

(4) $\sin^{-1}\left(\frac{3}{4}\right)$

19) What should be the minimum value of the force F so that block will remain at rest given that (μ = 0.2)



(1) 10 N

(2) 15 N

(3) 20 N

(4) 0 N

20) If $\sin\theta = \frac{12}{13}$ and θ lies in the second quadrant, then value of (cotθ + cosecθ) is :-

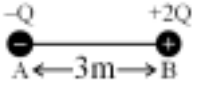
(1) $\frac{3}{2}$

(2) $\frac{2}{3}$

(3) $\frac{-3}{2}$

(4) $\frac{-2}{3}$

21) Charge 2Q and –Q are placed as shown in Fig. The point at which electric field intensity is zero will be :



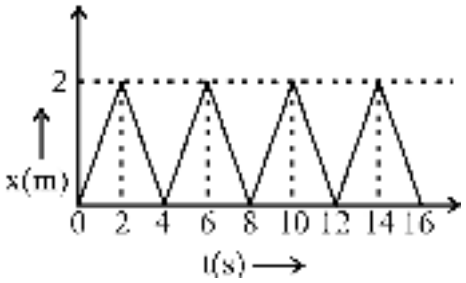
(1) somewhere between –Q and 2Q

(2) somewhere on the left of –Q at distance $\left(3\sqrt{2}+3\right)$.

(3) somewhere on the left of 2Q at distance $\left(3\sqrt{2}+6\right)$.

(4) Both (2) & (3) are correct

22) The figure shows the position-time (x-t) graph of one-dimensional motion of a body of mass 0.4 kg. The magnitude of each impulse is :-



(1) 0.4 N-s

(2) 0.8 N-s

(3) 1.6 N-s

(4) 0.2 N-s

23) An electric pump is used to fill an overhead tank of capacity 9 m^3 kept at a height of 10 m above the ground. If the pump takes 5 minutes of fill the tank by consuming 10 kW power, the efficiency of the pump should be : (Take g = 10 ms^2)

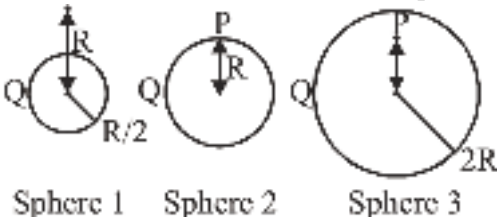
(1) 60%

(2) 40 %

(3) 20%

(4) 30 %

24) Charges Q is uniformly distributed in each of three dielectric solid spheres 1, 2 and 3 of radii R/2, R and 2R respectively, as shown in figure. If magnitudes of the electric fields at point P at a distance R from the centre of spheres 1, 2 and 3 are E_1, E_2 and E_3 respectively, then :-



(1) $E_1 > E_2 > E_3$

(2) $E_3 > E_1 > E_2$

(3) $E_1 = E_2 > E_3$

(4) $E_1 = E_2 = E_3$

25) A body A is projected upwards with a velocity of 98 m/s. The second body B is projected upwards with the same initial velocity but after 4 sec. Both the bodies will meet after :-

(1) 6 sec

(2) 8 sec

(3) 10 sec

(4) 12 sec

26) SI unit of potential gradient is :-

(1) V

(2) $V\ m^{-1}$

(3) V m

(4) m

27) When identical point charges are placed at the vertices of a cube of edge length a, each of them experiences a net force of magnitude F. Now these charges are placed on the vertices of another cube of edge length b. What will be magnitude of the net force on any of the charges ?
(These cubes are simply geometrical constructs and not made of any matter.)

(1) $\frac{a^2F}{b^2}$

(2) $\frac{b^2F}{a^2}$

(3) $\frac{aF}{b}$

(4) Insufficient information

28) A 100μF capacitor is charged to 200 volt. It is discharged through a 2 ohm resistance. The amount of heat generated will be :-

(1) 2 Joule

(2) 4 Joule

(3) 0.2 Joule

(4) 0.4 Joule

29) Consider a conductor of length 40 cm where a potential difference of 10 V is maintained between the ends of conductor. Find out the mobility of the electrons provided the drift velocity of the electrons is $5 \times 10^{-6}\ ms^{-1}$:-

(1) $8 \times 10^{-7}\ m^2\ V^{-1}\ s^{-1}$

(2) $2 \times 10^{-7}\ m^2\ V^{-1}\ s^{-1}$

(3) $4 \times 10^{-6}\ m^2\ V^{-1}\ s^{-1}$

(4) $10^{-7}\ m^2\ V^{-1}\ s^{-1}$

30) A boat moves relative to still water with a velocity which is n times the river flow velocity. At what angle to the stream direction must be boat move to cross the river along shortest path ?

(1) $\frac{\pi}{2}$

(2) $\sin^{-1}\left(\frac{1}{n}\right)$

(3) $\frac{\pi}{2} + \sin^{-1}\left(\frac{1}{n}\right)$

(4) $\frac{\pi}{2} - \sin^{-1}\left(\frac{1}{n}\right)$

31) If $\vec{a} = 3\hat{i} + 4\hat{j}$ & $\vec{b} = 7\hat{i} + 24\hat{j}$ then the vector having the same magnitude as $\vec{b}$ and same direction as $\vec{a}$ is:

(1) $15\hat{i} + 20\hat{j}$

(2) $10\hat{i} + 20\hat{j}$

(3) $20\hat{i} + 15\hat{j}$

(4) $15\hat{i} + 10\hat{j}$

32) **Assertion (A) :-** Two similarly charged bodies may attract each other. **Reason (R) :-** When charge on one body (Q) is much greater than that on another (q) and they are close enough to each other, then force of attraction between Q and induced charges exceeds the force of repulsion between Q and q.

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

(2) **(A)** is correct but **(R)** is not correct

(3) **(A)** is incorrect but **(R)** is correct

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

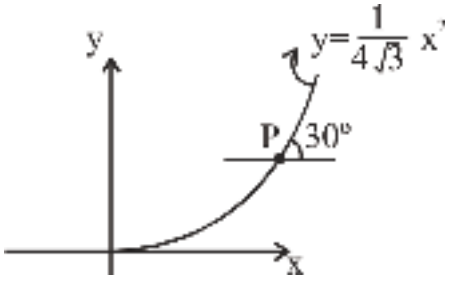
33) A particle is moving in a plane with velocity given by $u = u_0 i + (a\omega \cos\omega t)j$, where i and j are unit vectors along x and y axes respectively. If particle is at the origin at t = 0. Calculate the trajectory of the particle :-

(1) $y = a\sin\left(\frac{u_0}{\omega x}\right)$

(2) $y = a\sin\left(\frac{\omega x}{u_0}\right)$

(3) $y = \frac{1}{a}\sin\left(\frac{u_0}{\omega x}\right)$

(4) $y = \frac{1}{a}\sin\left(\frac{\omega x}{u_0}\right)$

34) 

If equation of the above curve is $y = \frac{1}{4\sqrt{3}}x^2$, then find co-ordinates of point P, at which curve makes angle 30º with x-axis.

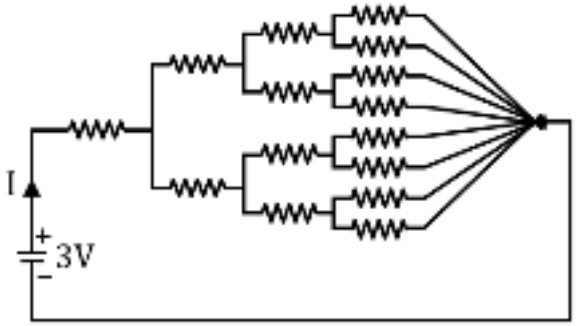
(1) $\left(1, \frac{1}{4\sqrt{3}}\right)$

(2) $\left(\sqrt{3}, \frac{1}{4\sqrt{3}}\right)$

(3) $\left(2, \frac{1}{\sqrt{3}}\right)$

(4) $\left(\sqrt{3}, \frac{\sqrt{3}}{4}\right)$

35) All resistances in figure are 1Ω each. The value of current 'I' is $\frac{a}{5}$ A. Then value of 'a' is _____.



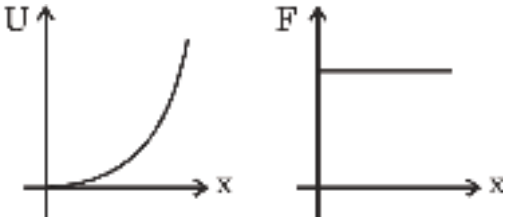
(1) 5

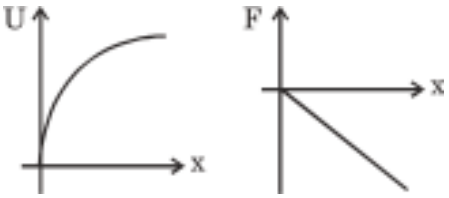
(2) 8

(3) 8/5

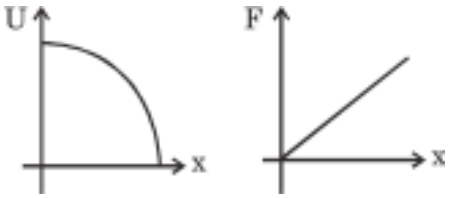
(4) 5/8

36) The potential energy function for a conservative force and its corresponding force curves are shown. Which of the following is correct ?

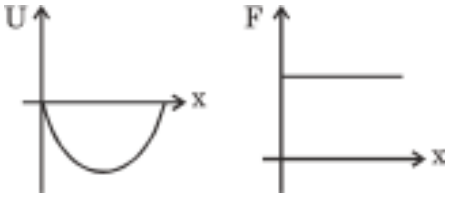
(1) 

(2) 

(3)



(4)

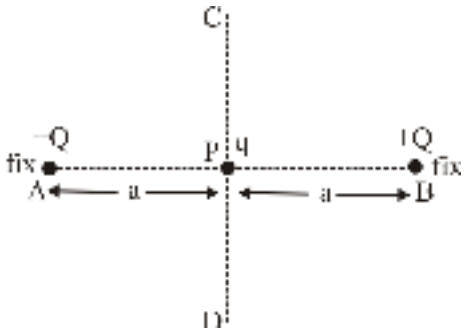


37) If K.E. increases by 4%. Then momentum will increase by :-

(1) 1.5%
(2) 9%
(3) 3%
(4) 2%

38)

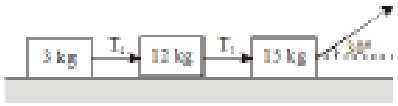
For the arrangement shown below, choose the correct statement :-



(1) If q is + ve, P is a position of stable equilibrium for displacement along AB
(2) If q is + ve, P is a position of unstable equilibrium for displacement along CD
(3) If q is – ve, P is a position of unstable equilibrium for displacement along AB
(4) All of the above

39)

The surfaces are frictionless, the ratio of T_1 and T_2 is:



(1) $\sqrt{3} : 2$
(2) $1 : \sqrt{3}$
(3) 1 : 5
(4) 5 : 1

40) A fully charged capacitor has a capacitance C. It is discharged through a small coil of resistance wire embedded in a thermally insulated block of specific heat capacity s and mass m. If the

temperature of the block is raised by ΔT, the potential difference V across the capacitance is :-

(1) $\sqrt{\frac{2mC\Delta T}{s}}$

(2) $\frac{mC\Delta T}{s}$

(3) $\frac{ms\Delta T}{C}$

(4) $\sqrt{\frac{2ms\Delta T}{C}}$

41) The velocity v of a moving particle varies with displacement as $x = \sqrt{v+1}$. The acceleration of the particle at x = 5 unit will be :-

(1) $\sqrt{6}$ unit

(2) 24 unit

(3) 240 unit

(4) 25 unit

42) The work done by external agent in stretching a spring of force constant k = 100 N/cm from deformation x_1 = 10 to deformation x_2 = 20 cm:-

(1) –150 J

(2) 50 J

(3) 150 J

(4) None of these

43) Two identical charged ball of same mass and charge, each of 'q' are suspended from a common point by equal strings, each of length '□'. In equilibrium distance between the balls is 'a' then tension in string is :-

(1) $\frac{q^2\ell}{4\pi\in_0 a^3}$

(2) $\frac{q^2\ell}{2\pi\in_0 a^3}$

(3) $\frac{q^2\ell}{\pi\in_0 a^2}$

(4) $\frac{q^2\ell}{4\pi\in_0 a^2}$

44) Consider a spherical symmetric distribution of charge. The centre of symmetry is at O. Electric flux linked with a spherical Gaussian surface of radius 'r' and centered at O is given by $\phi = \phi_0 r^4$. Volume charge density at a distance 'r' from O is given by :-

(1) $\frac{\in_0 \phi_0 r}{\pi}$

(2) $\frac{3\in_0\phi_0 r}{4}$

(3) $\frac{3\in_0\phi_0 r}{4\pi}$

(4) $\frac{\in_0\phi_0 r}{2\pi}$

45) A constant force F is applied on a body. The power (P) generated is related to the time elapsed (t) as

(1) $P \propto t^2$

(2) $P \propto t$

(3) $P \propto \sqrt{t}$

(4) $P \propto t^{3/2}$

CHEMISTRY

1) If 0.01 M solution of an electrolyte has a resistance of 40 ohms in a cell having a cell constant of 0.4 cm^{-1}, then its molar conductance in ohm^{-1} cm^2 mol^{-1} is :

(1) 10^2

(2) 10^4

(3) 10

(4) 10^3

2)

Electrolyte:	**KCl**	**KNO_3**	**HCl**	**NaOAc**	**NaCl**
Λ^∞ (S cm^2 mol^{-1}) :	149.9	145	426.2	91	126.5

Calculate Λ^∞_{HOAc} using appropriate molar conductances of the electrolytes listed above at infinite dilution in H_2O at 25°C :

(1) 217.5

(2) 390.7

(3) 552.7

(4) 517.2

3) How much chlorine will be liberated on passing one ampere current for 30 min. through NaCl solution?

(1) 0.66 mole

(2) 0.33 mole

(3) 0.66 g

(4) 0.33 g

4)

For the electrochemical cell, M | $[M^+$ || X^- | X, $E°(M^+/M)$ = 0.44 V and $E°$ (X/X^-) = 0.33 V.
From this data one can deduce that:

(1) $M + X \rightarrow M^+ + X^-$ is spontaneous reaction

(2) $M^+ + X^- \rightarrow M + X$ is spontaneous reaction

(3) E_{cell} = 0.77V

(4) E_{cell} = – 0.77V

5) Using same quantity of current, which among Na, Mg, and Al is deposited to the highest extent during electrolysis.

(1) Mg

(2) Al

(3) Na

(4) All in same amount

6) Aluminium oxide may be electrolysed at 1000°C to furnish aluminium metal (At. Mass = 27 amu; 1 Faraday = 96,500 Coulombs).
The cathode reaction is– $Al^{3+} + 3e^- \rightarrow Al$ To prepare 5.12 kg of aluminium metal by this method we require electricity of :

(1) 5.49×10^1 C

(2) 5.49×10^4 C

(3) 1.83×10^7 C

(4) 5.49×10^7 C

7) Which of the following will form a cell with the highest voltage?

(1) 0.1 M Ag^+, 2 MCo^{2+}

(2) 2 M Ag^+, 0.1 MCo^{2+}

(3) 1 M Ag^+, 1 M Co^{2+}

(4) 2 M Ag^+, 2 M Co^{2+}

8) What is the E^o_{cell} for the reaction
$Cu^{2+}(aq) + Sn^{2+}(aq) + Cu_5 + Sn^{4+}(aq)$ at 25°C if the equilibrium constant for the reaction is 1×10^6 ?

(1) 0.5328 V

(2) 0.3552 V

(3) 0.1773 V

(4) 0.7104 V

9) In electrolysis of dilute H_2SO_4, what is liberated at anode?

(1) H_2

(2) SO_4^{2-}

(3) SO_2

(4) O_2

10) The chemical reaction, $2AgCl_{(s)} + H_{2(g)} \rightarrow 2HCl_{(aq)} + 2Ag_{(s)}$ taking place in a galvanic cell is represented by the notation.

(1) $Pt_{(s)} \mid H_{2(g)}$,1 bar $\mid 1MKCl_{(aq)} \mid AgCl_{(s)} \mid Ag_{(s)}$

(2) $Pt_{(s)} \mid H_{2(g)}$,1 bar $\mid 1MHCl_{(aq)} \mid 1Mag^{+}_{(aq)} \mid Ag_{(s)}$

(3) $Pt_{(s)} \mid H_{2(g)}$,1 bar $\mid 1MHCl_{(aq)} \mid AgCl_{(s)} \mid Ag_{(s)}$

(4) $Pt_{(s)} \mid H_{2(g)}$,1 bar $\mid 1MHCl_{(aq)} \mid Ag_{(s)} \mid AgCl_{(s)}$

11) Specific conductance of 0.1 M HNO_3 is 6.3×10^{-2}. The molar conductance of the solution is :

(1) 100 ohm^{-1} cm^2

(2) 515 ohm^{-1} cm^2

(3) 630 ohm^{-1} cm^2

(4) 6300 ohm^{-1} cm^2

12) At 298K the standard free energy of formation of H_2O (□) is –237.20 kJ/mole while that of its ionisation into H^+ ion and hydroxyl ions is 80 kJ/mole, then the emf of the following cell at 298 K will be [Take Faraday constant F = 96500C]
$H_2(g, 1 \text{ bar}) \mid H^+ (1M) \mid\mid OH^- (1M) \mid O_2(g, 1 \text{ bar})$

(1) 0.40 V

(2) 0.81 V

(3) 1.23 V

(4) – 0.40 V

13) Which one of the following pairs of isomers and types of isomerism are correctly matched ?
(i) $[Co(NH_3)_5(NO_2)]Cl_2$ and $[Co(NH_3)_5(ONO)]Cl_2$ .....(Linkage)
(ii) $[Cu(NH_3)_4][PtCl_4]$ and $[Pt(NH_3)_4][CuCl_4]$ .....(Coordination)
(iii) $[PtCl_2(NH_3)_4]Br_2$ and $[PtBr_2(NH_3)_4]Cl_2$ .....(Ionization)
Select the correct answer using the codes given below :-

(1) (ii) and (iii)

(2) (i), (ii) and (iii)

(3) (i) and (iii)

(4) (i) and (ii)

14) The type of isomerism present in $[Cr(NH_3)_5 (NO_2)] Cl_2$ is :

(1) optical

(2) linkage

(3) hydrate

(4) polymerisation

15) Find the number of geometrical isomers of Ma_3b_3.

(1) 1
(2) 2
(3) 3
(4) 4

16) The Crystal Field splitting Energy (CFSE) for $[CoCl_6]^{4-}$ is 18000 cm^{-1}. The CFSE for $[CoCl_4]^{2-}$ will be-

(1) 6000 cm^{-1}
(2) 16000 cm^{-1}
(3) 18000 cm^{-1}
(4) 8000 cm^{-1}

17) Which of the following NOT conduct electricity :

(1) $[CrCl(H_2O)_5]Cl_2.H_2O$
(2) $[Pt(NH_3)_6]Cl_4$
(3) $[CrCl_3(H_2O)_3].3H_2O$
(4) $[Co(NH_3)_4]Cl_2$

18) **Assertion :-** EDTA ligand is hexadentate and chelating ligand.
Reason :- EDTA have 4 'N' and 2 'O' donor atoms.

(1) Both Assertion & Reason are True but Reason is not correct explanation of the Assertion.
(2) Assertion is True but the Reason is False.
(3) Assertion is False but the Reason is True.
(4) Both Assertion & Reason are True but Reason is correct explanation of the Assertion.

19) Which of the following complexes has a geometry different from others?

(1) $[NiCl_4]^{2-}$
(2) $Ni(CO)_4$
(3) $[Ni(CN)_4]^{2-}$
(4) $[Zn(NH_3)_4]^{2+}$

20) In octahedral complex central metal ion is Fe^{+3}. In presence of strong field ligand it shows paramagnetic character due to presence of :-

(1) 5 unpaired e^-
(2) 3 unpaired e^-
(3) 2 unpaired e^-

(4) 1 unpaired e^-

21) Which of the following complex is colourless :-

(1) $[Cr(H_2O)_6]Cl_3$
(2) $[Sc(H_2O)_6]Cl_3$
(3) $[Ti(H_2O)_6]^{3+}$
(4) $KMnO_4$

22) Most stable complex ion is :-

(1) $[FeCl_6]^{-3}$
(2) $[Fe(CN)_6]^{-3}$
(3) $[FeF_6]^{3-}$
(4) $[Fe(H_2O)_6]^{+3}$

23) Predict the incorrect statement about $[Co(NH_3)_6]^{+3}$

(1) It is low spin complex
(2) Its hybridization is sp^3d^2
(3) It is diamagnetic in nature
(4) Its geometry is octahedral

24) Which of the following is correctly matched :-

	Column-I	Column-II	Column-III
(A)	$[Cr(CO)_6]$	Paramagnetic	Octahedral, sp^3d^2
(B)	$[Fe(CO)_5]$	Paramagnetic	Trigonal bipyramidal, sp^3d
(C)	$[Co(CO)_4]^-$	Diamagnetic	Tetrahedral, sp^3
(D)	$[Ni(CO)_4]$	Diamagnetic	Square planar, dsp^2

(1) A
(2) B
(3) C
(4) D

25) The IUPAC name for the complex, $[Co(NH_3)_2Cl(ONO)(Py)_2]NO_3$ is ?

(1) Diamminechlorido dipyridinenitrito-N-cobaltate (III) nitrate
(2) Diammine chloridonitrito-O-bipyridine cobalt (III) nitrate
(3) Diamminechloridonitrito-O-dipyridine cobalt (III) nitrate
(4) Chloridodiammineitrito-O-bis (pyridine) cobalt (III) nitrate

26) Find the incorrect relation

(1) SCN^- → Ambidentate ligand

(2) $N(CH_2CH_2NH_2)_3$ → polydentate ligand

(3) H_2O → Uni dentate ligand

(4) $EDTA^{4-}$ → didentate ligand

27) Meso tartaric acid & d-tartaric acid are-

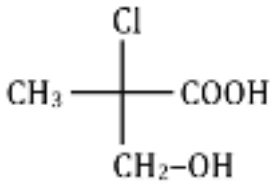
(1) Enantiomers

(2) Diastereomers

(3) Homomer

(4) Chain isomer

28) Configuration of following chiral-C is

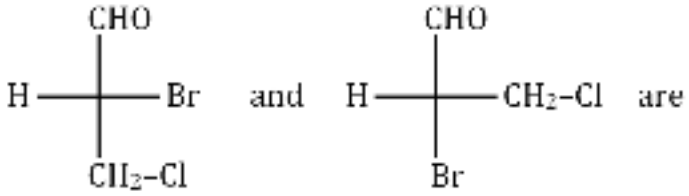


(1) R

(2) S

(3) d

(4)

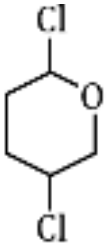
29) 

(1) Enantiomers

(2) Diastereomers

(3) Homomer

(4) Position isomers

30) **Assertion :** 

can exist in meso.

Reason : It can have plane of symmetry.

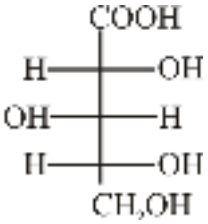
(1) If both Assertion and Reason are True and the Reason is a correct explanation of the Assertion.

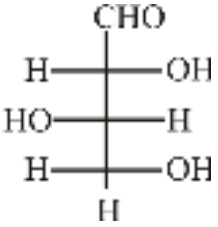
(2) If both Assertion and Reason are True but Reason is not a correct explanation of the Assertion.

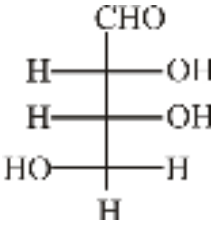
(3) If Assertion is True but the Reason is False.

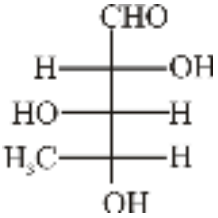
(4) If both Assertion and Reason are False.

31) Which of the following compound show L-configuration :-

(1) 

(2) 

(3) 

(4) 

32) Compound 'A' and Compound 'B' have same structure and configuration of compound 'A' and compound 'B' is 2R, 3S and 2S, 3S respectively. Correct statement about compound 'A' and compound 'B' is

(1) 'A' and 'B' are diastereomers.

(2) 'A' and 'B' are enantiomers if both are active.

(3) 'A' and 'B' are identical if both are inactive.

(4) 'A' and 'B' are functional isomers.

33)

Given below are two statements, one is labelled as Assertion A and the other is labelled as Reason R.
Assertion A : When 2-chlorobutane is heated with Alc. KOH then trans-2-butene is major product.
Reason R : 2-Butene is more stable than 1-Butene.
In the light of the above statements, choose the most appropriate answer from the options given below.

(1) Both assertion and reason are true and the reason is a correct explanation of the assertion.

(2) Both assertion and reason are true and the reason is not a correct explanation of the assertion.

(3) Assertion is true but the reason is false.

(4) Assertion is false but reason is true.

34) Given below are two statements:
Statement-I : When Alkyl halide is reacted with AgCN then major product is Alkyl isocyanide.
Statement-II : When alkyl halide is reacted with $NaNO_2$ then nitro alkane ($R—NO_2$) is major product.
In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both statement-I and statement-II are correct.

(2) Statement-I is correct but statement-II is incorrect.

(3) Both statement-I and statement-II are incorrect.

(4) Statement-I is incorrect but statement-II is correct.

35) Given below are two statements, one is labelled as Assertion A and the other is labelled as Reason R.
Assertion A : When alkyl halide is reacted with Na/dry ether then product is alkane while when reacted with Mg/dry ether then product is Grignard reagent.
Reason R : Alkane is non polar while Grignard reagent is highly polar.
In the light of the above statements, choose the most appropriate answer from the options given below.

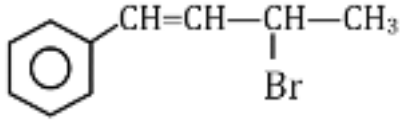
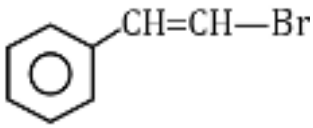
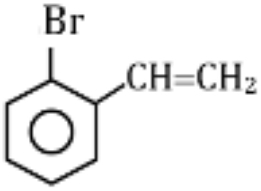
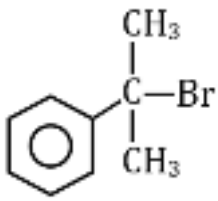
(1) Both assertion and reason are true and the reason is a correct explanation of the assertion.

(2) Both assertion and reason are true and the reason is not a correct explanation of the assertion.

(3) Assertion is true but the reason is false.

(4) Assertion is false but reason is true.

36)

	Column-I (Halide)		**Column-II (Type)**
(A)		(P)	Aryl halide
(B)		(Q)	Benzylic halide
(C)		(R)	Allylic halide
(D)		(S)	Vinylic halide

Correct match is-

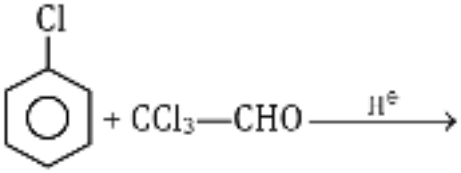
(1) A → R, B → S, C → P, D → Q

(2) A → Q, B → S, C → P, D → R

(3) A → P, B → Q, C → R, D → S

(4) A → S, B → R, C → P, D → Q

37)

	Column-I		Column-II
(A)		(P)	Phosgene gas
(B)	$CHCl_3 + \frac{1}{2}O_2 \rightarrow$	(Q)	DDT
(C)	$CH_4 + \underset{(excess)}{Cl_2} \xrightarrow{h\nu}$	(R)	Fire exfinguisher

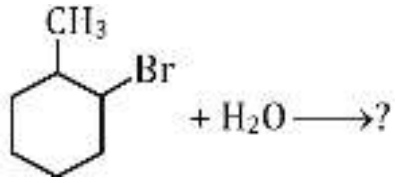
Correct match is-

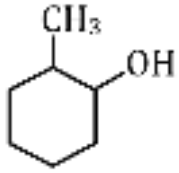
(1) A → P, B → Q, C → R

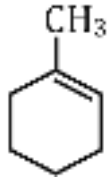
(2) A → Q, B → P, C → R

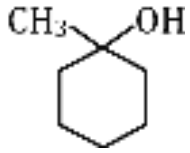
(3) A → R, B → Q, C → P

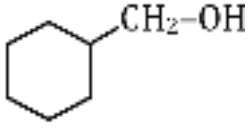
(4) A → P, B → R, C → Q

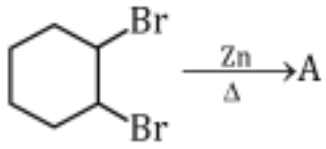
38) 

(1) 

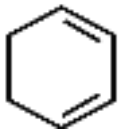
(2) 

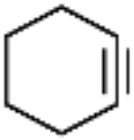
(3) 

(4) 

39) 

Product 'A' is

(1) 

(2) 

(3) 

(4) 

(a) $CH_3-\underset{\underset{Br}{|}}{CH}-CH_3 + CH_3-CH_2-\bar{O}\,\overset{+}{Na} \longrightarrow$

(b) $CH_3-CH_2-CH_2-Br + CH_3-CH_2-\bar{O}\,\overset{+}{Na} \xrightarrow{\Delta}$

(c) $CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_2-Br + CH_3-\bar{O}\,\overset{+}{Na} \longrightarrow$

(d) $CH_3-CH_2-CH_2-Br + CH_3-\bar{O}\,\overset{+}{Na} \longrightarrow$

(e) $CH_3-Br + CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-\bar{O}\,\overset{+}{Na} \longrightarrow$

40)

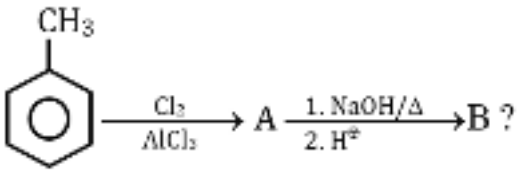
Out of above reactions the total reactions which gives ether as a major product

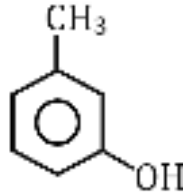
(1) 3

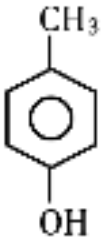
(2) 2

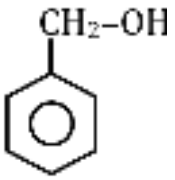
(3) 4

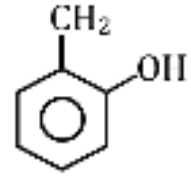
(4) 5

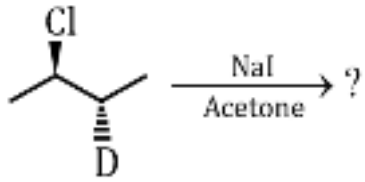
41) 

(1) 

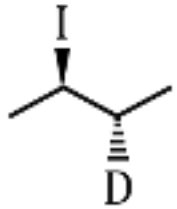
(2) 

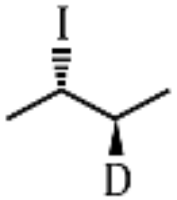
(3) 

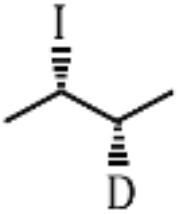
(4) 

42) 

(1) 

(2) 

(3) 

(4) 

43) The correct statement regarding the following two reactions.

(A) $CH_3 - CH_2 - Cl \xrightarrow{\bar{O}H}$

(B) $CH_3 - CH_2 - Cl \xrightarrow{\bar{S}H}$

(1) $\overset{\Theta}{O}H$ is better nucleophile than $\overset{\Theta}{S}H$

(2) A is faster than B

(3) Both reaction follows S_N2 path

(4) Major product in both reactions is ethane

44) In which reaction halide is not obtained

(a) $CH_3 - CH_2 - CH_2 - OH \xrightarrow{H-Cl}$

(b) $CH_3 - CH_2 - CH_2 - OH \xrightarrow{H-Br}$

(c) $CH_3 - CH_2 - CH_2 - OH \xrightarrow{H-I}$

(d) $Ph - OH \xrightarrow{H-Br}$

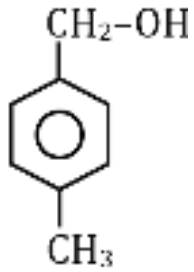
(1) a, b

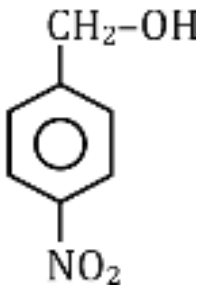
(2) a, d

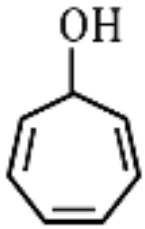
(3) a, b, d

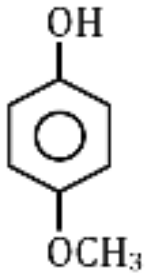
(4) All

45) Which is maximum reactive towards H–Br

(1) 

(2) 

(3) 

(4) 

BIOLOGY

1) Which of the following correctly represents the sequence of structures through which sperm passes after leaving the seminiferous tubules?

(1) Rete testis → vasa efferentia → epididymis → vas deferens
(2) Epididymis → rete testis → vasa efferentia → vas deferens
(3) Rete testis → epididymis → vasa efferentia → vas deferens
(4) Vasa efferentia → rete testis → epididymis → vas deferens

2) Which cells are primarily responsible for providing nutrition to developing germ cells in the seminiferous tubules?

(1) Leydig cells
(2) Sertoli cells
(3) Spermatogonia
(4) Primary spermatocytes

3) **Assertion:** Menstruation occurs when fertilisation does not take place.
Reason: In the absence of fertilisation, the corpus luteum degenerates and progesterone levels decline.

(1) Both Assertion and Reason are true, and Reason is the correct explanation.
(2) Both Assertion and Reason are true, but Reason is not the correct explanation.
(3) Assertion is true, but Reason is false.
(4) Assertion is false, but Reason is true.

4) Leydig cells are located :

(1) Inside seminiferous tubules
(2) Between seminiferous tubules
(3) Inside epididymis
(4) Inside prostate gland

5) A primary spermatocyte in humans :

(1) Haploid and undergoes mitosis
(2) Diploid and undergoes mitosis

(3) Haploid and undergoes meiosis II
(4) Diploid and undergoes meiosis I

6) One primary spermatocyte ultimately produces:

(1) 2 spermatozoa
(2) 3 spermatozoa
(3) 4 spermatozoa
(4) 8 spermatozoa

7) **Assertion:** FSH is responsible for ovulation.
Reason: A sudden increase in LH causes ovulation.

(1) Both Assertion and Reason are true, and Reason is the correct explanation.
(2) Both Assertion and Reason are true, but Reason is not the correct explanation.
(3) Assertion is true, but Reason is false.
(4) Assertion is false, but Reason is true.

8) Match the structure with its correct function.

Structure	Function
A. Acrosome	1. Sperm movement
B. Middle piece	2. Enzymes for sperm entry
C. Tail	3. Energy production
D. Endometrium	4. Implantation

(1) A-3, B-1, C-2, D-4
(2) A-2, B-3, C-1, D-4
(3) A-2, B-1, C-3, D-4
(4) A-4, B-3, C-1, D-2

9)

Which statement about spermiogenesis is correct?

(1) Spermatogonia transform into primary spermatocytes
(2) Spermatids transform into spermatozoa
(3) Spermatozoa undergo meiosis
(4) Secondary spermatocytes transform into spermatids through mitosis

10) The acrosome of sperm is derived from :

(1) Mitochondria
(2) Golgi apparatus
(3) Endoplasmic reticulum
(4) Nucleus

11) Which part of sperm contains numerous mitochondria?

(1) Head
(2) Neck
(3) Middle piece
(4) Tail end

12) Which hormone directly stimulates Leydig cells to produce testosterone?

(1) FSH
(2) LH
(3) Progesterone
(4) Estrogen

13) Which of the following is not a function of the placenta?

(1) Exchange of nutrients
(2) Exchange of gases
(3) Removal of fetal wastes
(4) Production of sperm

14) The usual site of fertilisation in humans is:

(1) Cervix
(2) Uterus
(3) Ampullary-isthmic junction of fallopian tube
(4) Vagina

15) Which layer of the uterus undergoes cyclic changes during the menstrual cycle?

(1) Perimetrium
(2) Myometrium
(3) Endometrium
(4) Serosa

16) Ovulation in a typical 28-day menstrual cycle occurs approximately on:

(1) Day 1
(2) Day 7
(3) Day 14
(4) Day 28

17) The LH surge causes:

(1) Menstruation
(2) Ovulation

(3) Implantation

(4) Fertilisation

18) Which hormone maintains the endometrium during the luteal phase?

(1) FSH

(2) LH

(3) Progesterone

(4) Oxytocin

19) **Assertion:** The corpus luteum is important for maintaining pregnancy during the early stages.
Reason: The corpus luteum secretes progesterone.

(1) Both Assertion and Reason are true, and Reason is the correct explanation.

(2) Both Assertion and Reason are true, but Reason is not the correct explanation.

(3) Assertion is true, but Reason is false.

(4) Assertion is false, but Reason is true.

20) Consider the following statements:
A. Spermatogenesis begins at puberty.
B. Spermatogonia are diploid.
C. Secondary spermatocytes are diploid.
D. Spermatids are haploid.
Which statements are correct?

(1) A, B and D only

(2) A and C only

(3) B and D only

(4) A, B, C and D

21) Consider the following statements regarding the menstrual cycle:
A. FSH stimulates follicular growth.
B. Growing follicles secrete estrogen.
C. LH surge causes ovulation.
D. Corpus luteum primarily secretes testosterone.
Which are correct?

(1) A, B, C and D

(2) A and D

(3) B and D

(4) A, B and C

22) Which sequence correctly represents early embryonic development?

(1) Zygote → Blastocyst → Morula → Implantation

(2) Morula → Zygote → Blastocyst → Implantation

(3) Zygote → Morula → Blastocyst → Implantation

(4) Zygote → Implantation → Morula → Blastocyst

23) **Assertion:** Implantation occurs in the ovary.
Reason: The blastocyst attaches to the endometrium of the uterus.

(1) Both Assertion and Reason are true, and Reason is the correct explanation.

(2) Both Assertion and Reason are true, but Reason is not the correct explanation.

(3) Assertion is true, but Reason is false.

(4) Assertion is false, but Reason is true.

24) Match Column I with Column II.

Column I	Column II
A. Sertoli cells	1. Testosterone
B. Leydig cells	2. Nutrition/support to germ cells
C. Corpus luteum	3. Progesterone
D. Placenta	4. hCG

(1) A-1, B-2, C-4, D-3

(2) A-3, B-1, C-2, D-4

(3) A-2, B-3, C-1, D-4

(4) A-2, B-1, C-3, D-4

25) Arrange the following events in the correct order:
1. Ovulation
2. Formation of corpus luteum
3. LH surge
4. Development of dominant follicle

(1) 4 → 3 → 1 → 2

(2) 3 → 4 → 1 → 2

(3) 4 → 1 → 3 → 2

(4) 1 → 3 → 4 → 2

26) Arrange the following in the correct order of sperm formation :

(1) Spermatogonia → Primary spermatocyte → Secondary spermatocyte → Spermatid → Spermatozoa

(2) Spermatogonia → Secondary spermatocyte → Primary spermatocyte → Spermatid → Spermatozoa

(3) Primary spermatocyte → Spermatogonia → Secondary spermatocyte → Spermatozoa → Spermatid

(4) Spermatid → Spermatogonia → Primary spermatocyte → Secondary spermatocyte → Spermatozoa

27) A woman has a normal 28-day menstrual cycle. If fertilisation does not occur, which hormonal change is most directly associated with menstruation?

(1) Increased progesterone secretion

(2) Decreased progesterone secretion

(3) Increased hCG secretion

(4) Increased testosterone secretion

28) Which combination is correctly associated with the placenta?

(1) hCG, hPL, estrogen and progesterone

(2) FSH, LH, testosterone and insulin

(3) TSH, ADH, oxytocin and glucagon

(4) FSH, hCG, testosterone and thyroxine

29) During spermatogenesis, reduction in chromosome number occurs during :

(1) Spermatogonia → primary spermatocyte

(2) Primary spermatocyte → secondary spermatocyte

(3) Secondary spermatocyte → spermatid

(4) Spermatid → spermatozoa

30) A female undergoes fertilisation normally, but implantation does not occur. Which of the following is most likely to happen?

(1) hCG secretion will maintain the corpus luteum

(2) Corpus luteum will continue secreting progesterone indefinitely

(3) Corpus luteum will degenerate and progesterone level will fall

(4) LH surge will continue throughout pregnancy

31) In a double-stranded B-DNA, the pitch of the helix is 3.4nm (34 AA). What is the distance between two consecutive base pairs?

(1) 3.4 nm

(2) 0.34 nm

(3) 34 nm

(4) 0.034 nm

32) **Statement 1:** Histones are rich in the basic amino acid residues lysine and arginine, which carry positive charges in their side chains.
Statement 2: A typical nucleosome contains 200 bp of DNA helix wrapped around a negatively charged histone octamer.

(1) Both Statement 1 and Statement 2 are correct.

(2) Both Statement 1 and Statement 2 are incorrect.

(3) Statement 1 is correct but Statement 2 is incorrect.

(4) Statement 1 is incorrect but Statement 2 is correct.

33)

In their landmark 1952 experiment proving DNA as the genetic material, Hershey and Chase used radioactive isotopes to label bacteriophage components. Which isotope was used to label phage DNA?

(1) ^{35}S

(2) ^{15}N

(3) ^{32}P

(4) ^{14}C

34) **Assertion (A):** RNA is chemically less stable and more reactive than DNA.
Reason (R): Every nucleotide in RNA has an additional 2'-OH group present in its ribose sugar.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) (A) is false but (R) is true.

35) Who proposed the Central Dogma in molecular biology, which states that genetic information flows from DNA → RNA → Protein?

(1) James Watson

(2) Francis Crick

(3) Maurice Wilkins

(4) Erwin Chargaff

36) Match Column I with Column II regarding eukaryotic RNA polymerases and select the correct option:

Column I (Enzyme)	**Column II (Transcribed Product)**
A. RNA Polymerase I	**1.** hnRNA (precursor of mRNA)
B. RNA Polymerase II	**2.** tRNA, *5*SrRNA, and snRNAs
C. RNA Polymerase III	**3.** 28S, 18S, and 5.8SrRNAs

(1) A-3, B-1, C-2

(2) A-1, B-2, C-3

(3) A-2, B-3, C-1

(4) A-3, B-2, C-1

37) During post-transcriptional processing of eukaryotic pre-mRNA, capping involves the addition of an unusual nucleotide at the 5'-end. What is this nucleotide?

(1) Adenylate residues

(2) Methyl guanosine triphosphate

(3) Deoxythymidine monophosphate

(4) Cytidine diphosphate

38) Consider the following statements regarding the properties of the genetic code :

1. The codon is triplet, and 61codons code for amino acids while 3 codons function as stop codons.
2. The genetic code is unambiguous, meaning one codon codes for only one specific amino acid.
3. The genetic code is degenerate because all amino acids are coded by only a single codon.
4. AUG has a dual function: it codes for Methionine and also acts as an initiator codon.
How many of the above statements are correct?

(1) Only one
(2) Only two
(3) Only three
(4) All four

39) According to the Human Genome Project (HGP), which is the largest known human gene?

(1) Dystrophin
(2) Insulin
(3) Hemoglobin β-chain
(4) Tumor suppressor p53

40) Read the following statements regarding the structural genes of the Lac Operon:
(a) The z gene codes for β-galactosidase, which hydrolyzes lactose into glucose and galactose.
(b) The y gene codes for permease, which increases the permeability of the cell to β-galactosides.
(c) The a gene codes for transacetylase.
(d) All three structural genes (z, y, a) are transcribed together as a single polycistronic mRNA.
Which of the above statements are correct ?

(1) (a) and (b) only
(2) (c) and (d) only
(3) (a), (b), and (c) only
(4) (a), (b), (c), and (d)

41) Who postulated the existence of an 'adapter molecule' (later identified as tRNA) that reads the genetic code and binds to specific amino acids?

(1) James Watson
(2) Francis Crick
(3) Har Gobind Khorana
(4) Marshall Nirenberg

42) **Statement 1:** Meselson and Stahl proved the semi-conservative replication of DNA using the heavy isotope ^{15}N in *Escherichia coli*.
Statement 2: Cesium chloride (CsCl) density gradient centrifugation was used to separate heavy ^{15}N-DNA from normal ^{14}N-DNA.

(1) Both Statement 1 and Statement 2 are correct.
(2) Both Statement 1 and Statement 2 are incorrect.
(3) Statement 1 is correct but Statement 2 is incorrect.
(4) Statement 1 is incorrect but Statement 2 is correct.

43) Which of the following codons acts as the primary initiation codon during translation in protein synthesis?

(1) UAA

(2) UAG

(3) AUG

(4) UGA

44) Match Column I with Column II regarding salient features of the Human Genome Project:

Column I (Chromosome / Feature)	**Column II (Number / Value)**
A. Total base pairs in human genome	**1.** 2968 genes
B. Chromosome 1	**2.** 231 genes
C. Y Chromosome	**3.** *3.1647* billion(*3×109* bp)

(1) A-1, B-2, C-3

(2) A-2, B-3, C-1

(3) A-3, B-2, C-1

(4) A-3, B-1, C-2

45) In DNA fingerprinting, the technique relies on identifying repetitive DNA sequences that show high degree of polymorphism. What are these short repetitive sequences called?

(1) Expressed Sequence Tags (ESTs)

(2) Variable Number of Tandem Repeats (VNTRs)

(3) Single Nucleotide Polymorphisms (SNPs)

(4) Okazaki fragments

46) In Frederick Griffith's transformation experiment, which strain of *Streptococcus pneumoniae* was virulent and possessed a smooth mucous (polysaccharide) coat?

(1) S strain

(2) R strain

(3) Both S and R strains

(4) Heat-killed R strain

47) **Statement 1:** DNA replication is continuous on the template strand with polarity 3' → 5'.
Statement 2: The discontinuously synthesized DNA fragments on the lagging strand are joined by the enzyme DNA ligase.

(1) Both Statement 1 and Statement 2 are incorrect.

(2) Both Statement 1 and Statement 2 are correct.

(3) Statement 1 is correct but Statement 2 is incorrect.

(4) Statement 1 is incorrect but Statement 2 is correct.

48) Which enzyme acts as a ribozyme and catalyzes peptide bond formation during translation in prokaryotes?

(1) 28S rRNA

(2) 16S rRNA

(3) 5S rRNA

(4) 23S rRNA

49) **Assertion (A):** Charging of tRNA (aminoacylation) is an essential initial step before translation begins.
Reason (R): Formation of a peptide bond between two amino acids requires energy, and linked amino acids are brought into close proximity by tRNA molecules on the ribosome.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) (A) is false but (R) is true.

50) What is the inducer molecule that binds to the repressor protein and turns on the expression of the *lac* operon in *E. coli*?

(1) Glucose

(2) Galactose

(3) Lactose

(4) Sucrose

51) Consider the following statements regarding DNA structure and Chargaff's rules:
1. Chargaff's rules state that in double-stranded DNA, the ratio between Adenine and Thymine, and Guanine and Cytosine are constant and equal to one.
2. The two polynucleotide chains of a DNA double helix have anti-parallel polarity.
3. Adenine forms three hydrogen bonds with Thymine, while Guanine forms two hydrogen bonds with Cytosine.
4. The backbone of a DNA strand is constituted by sugar-phosphate chains.
How many of the above statements are correct?

(1) Only one

(2) Only two

(3) Only three

(4) All four

52) Who developed the technique of DNA fingerprinting using Variable Number of Tandem Repeats (VNTRs) as probes?

(1) Alec Jeffreys

(2) Francis Crick

(3) Kary Mullis

(4) Francois Jacob

53) Match Column I (Scientist) with Column II (Key Discovery / Experiment) and select the correct option:

Column I	Column II
A. Avery, MacLeod, and McCarty	1. Unambiguous proof that DNA is the genetic material
B. Hershey and Chase	2. Biochemical characterization of Transforming Principle
C. Jacob and Monod	3. Deciphered genetic code using cell-free system
D. Marshall Nirenberg	4. Operon model of gene regulation

(1) A-2, B-1, C-4, D-3

(2) A-1, B-2, C-3, D-4

(3) A-2, B-4, C-1, D-3

(4) A-4, B-1, C-2, D-3

54) In a transcription unit, the strand that has the polarity 5' → 3' and has the same sequence as the RNA transcript (except thymine in place of uracil) is referred to as the :

(1) Template strand

(2) Coding strand

(3) Non-coding strand

(4) Anti-sense strand

55) Read the following statements regarding the methodologies of the Human Genome Project (HGP):
(a) Identifying all the genes that are expressed as RNA is referred to as Expressed Sequence Tags (ESTs).
(b) Blind sequencing of the entire genome containing coding and non-coding sequences is called Sequence Annotation.
(c) For sequencing, total DNA from a cell is isolated, converted into random fragments, and cloned into suitable host vectors like BAC and YAC.
(d) Automated DNA sequencers used in HGP worked on the principle of a method developed by Frederick Sanger.
Which of the above statements are correct ?

(1) (a) and (b) only

(2) (c) and (d) only

(3) (a), (b), and (c) only

(4) (a), (b), (c), and (d)

56) In his classical breeding experiments on garden peas (*Pisum sativum*), how many pairs of contrasting traits were studied by Gregor Mendel?

(1) 7

(2) 14

(3) 4

(4) 12

57) **Statement 1:** According to Mendel's Law of Segregation, alleles of a pair segregate from each other such that a gamete receives only one of the two factors.
Statement 2: The Law of Segregation is universal and shows no exceptions in sexually reproducing diploid organisms.

(1) Both Statement 1 and Statement 2 are correct.

(2) Both Statement 1 and Statement 2 are incorrect.

(3) Statement 1 is correct but Statement 2 is incorrect.

(4) Statement 1 is incorrect but Statement 2 is correct.

58) When a cross is made between a tall pea plant (Tt) and a dwarf pea plant (tt), what will be the phenotypic ratio of the offspring?

(1) 3 : 1

(2) 1 : 1

(3) 9 : 3 : 3 : 1

(4) 1 : 2 : 1

59) **Assertion (A):** In ABO blood grouping in humans, the allele I^A and I^B are codominant over each other.
Reason (R): When both I^A and I^B are present together in an individual, both express their own types of sugars on the surface of red blood cells.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) (A) is false but (R) is true.

60) In snapdragon (*Antirrhinum majus*), a cross between a red-flowered plant (RR) and a white-flowered plant (rr) produces pink-flowered offspring (Rr). What will be the genotypic ratio in the F_2 generation?

(1) 3 : 1

(2) 1 : 2 : 1

(3) 9 : 3 : 3 : 1

(4) 2 : 1 : 1

61) Match Column I with Column II regarding genetic disorders and select the correct option:

Column I (Disorder)	**Column II (Nature of Inheritance)**
A. Sickle-cell anemia	1. Sex-linked recessive disorder
B. Haemophilia	2. Autosomal recessive disorder
C. Myotonic dystrophy	3. Autosomal dominant disorder
D. Klinefelter's syndrome	4. Chromosomal disorder (*44+XXY*)

(1) A-1, B-2, C-4, D-3

(2) A-3, B-1, C-2, D-4

(3) A-2, B-3, C-1, D-4

(4) A-2, B-1, C-3, D-4

62) In a dihybrid cross between pea plants heterozygous for both round seeds/yellow cotyledons (RrYy) and double recessive wrinkled seeds/green cotyledons (rryy), the proportion of offspring

showing recombinant phenotypes will be:

(1) 25%
(2) 50%
(3) 75%
(4) 100%

63) Consider the following statements regarding sex determination in organisms:
1. In humans and *Drosophila*, males are heterogametic (XY) while females are homogametic (XX).
2. In birds, females are heterogametic (□□) while males are homogametic (□□).
3. In grasshoppers, male heterogamety is of the □□ type where males have one less chromosome than females.
4. Honeybees exhibit haplodiploid sex determination where males develop parthenogenetically from unfertilized eggs.
How many of the above statements are correct?

(1) Only one
(2) Only two
(3) Only three
(4) All four

64) Which chromosomal condition is characterized by the absence of one of the X chromosomes in human females, leading to a karyotype of 45 with □□?

(1) Down's syndrome
(2) Turner's syndrome
(3) Klinefelter's syndrome
(4) Edward's syndrome

65) Read the following statements regarding linkage and recombination studied by T.H. Morgan in *Drosophila*:
(a) Genes located on the same chromosome that are tightly linked show very low recombination frequencies.
(b) Alfred Sturtevant used the frequency of recombination between gene pairs on the same chromosome as a measure of the distance between genes.
(c) Tightly linked genes for yellow body and white eye showed 1.3% recombination.
(d) Loosely linked genes for white eye and miniature wing showed 37.2% recombination.
Which of the above statements are correct?

(1) (a) and (b) only
(2) (c) and (d) only
(3) (a), (b), and (c) only
(4) (a), (b), (c), and (d)

66) In human sickle-cell anemia, the defect is caused by the substitution of Glutamic acid by which amino acid at the sixth position of the beta-globin chain?

(1) Valine

(2) Glycine

(3) Lysine

(4) Leucine

67) **Statement 1:** Phenylketonuria is an inborn error of metabolism inherited as an autosomal recessive trait.
Statement 2: Affected individuals lack the hepatic enzyme phenylalanine hydroxylase, leading to accumulation of phenylalanine and its conversion to phenylpyruvic acid.

(1) Both Statement 1 and Statement 2 are correct.

(2) Both Statement 1 and Statement 2 are incorrect.

(3) Statement 1 is correct but Statement 2 is incorrect.

(4) Statement 1 is incorrect but Statement 2 is correct.

68) How many types of gametes will be produced by an individual with the genotype □□□□□□?

(1) 4

(2) 8

(3) 16

(4) 32

69) Match Column I with Column II regarding genetic discoveries/concepts and select the correct option:

Column I (Concept / Term)	**Column II (Scientist / Example)**
A. Chromosomal Theory of Inheritance	1. Henking
B. Discovery of X-body	2. Sutton and Boveri
C. Polygenic inheritance	3. Human skin color
D. Pleiotropy	4. Phenylketonuria

(1) A-1, B-2, C-4, D-3

(2) A-2, B-3, C-1, D-4

(3) A-2, B-1, C-3, D-4

(4) A-4, B-1, C-2, D-3

70) Which genetic disorder is caused by the presence of an additional copy of chromosome number 21 (Trisomy 21)?

(1) Turner's syndrome

(2) Down's syndrome

(3) Klinefelter's syndrome

(4) Thalassemia

71) **Statement 1 :** Thalassemia is an autosome-linked recessive blood disease transmitted from carrier parents to offspring.
Statement 2 : α-Thalassemia is controlled by two closely linked genes HBA1 and HBA2 on

chromosome 16.

(1) Both Statement 1 and Statement 2 are correct.

(2) Both Statement 1 and Statement 2 are incorrect.

(3) Statement 1 is correct but Statement 2 is incorrect.

(4) Statement 1 is incorrect but Statement 2 is correct.

72) A male with blood group AB marries female with blood group 'O'. What are the possible blood groups of their biological children?

(1) Only O

(2) A and B only

(3) A, B and AB only

(4) A, B, AB, and O

73) Consider the following statements regarding Sickle-cell anemia:
1. The disease is controlled by a single pair of alleles, Hb^A and Hb^S.
2. Homozygous individuals (Hb^SHb^S) show the diseased phenotype.
3. Heterozygous individuals (Hb^AHb^S) are unaffected carriers but show sickling under low oxygen tension.
4. It is a qualitative problem of synthesizing an incorrectly functioning globin molecule.
How many of the above statements are correct?

(1) Only one

(2) Only two

(3) Only three

(4) All four

74) What percentage of plants in the F_2 generation of a Mendelian monohybrid cross are homozygous for the trait?

(1) 25%

(2) 50%

(3) 75%

(4) 100%

75) **Assertion (A):** In honeybees, males do not have a father and cannot have sons, but they have a grandfather and can have grandsons.
Reason (R): Male honeybees (drones) develop parthenogenetically from unfertilized eggs produced by the queen bee.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is notT the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) (A) is false but (R) is true.

76) Which of the following traits in garden pea was studied by Mendel as a recessive trait?

(1) Yellow pod color

(2) Axial flower position

(3) Round seed shape

(4) Tall stem height

77) Read the following statements regarding pedigree analysis:
(a) In pedigree charts, solid/filled symbols represent affected individuals.
(b) A square symbol represents a male, while a circle represents a female.
(c) Mating between relatives is represented by a double line connecting symbols.
(d) Myotonic dystrophy is an example of an autosomal dominant trait traceable through pedigree.
Which of the above statements are correct ?

(1) (a) and (b) only

(2) (c) and (d) only

(3) (a), (b), and (c) only

(4) (a), (b), (c), and (d)

78) Failure of segregation of chromatids during cell division cycle results in the gain or loss of a chromosome(s), called:

(1) Aneuploidy

(2) Polyploidy

(3) Translocation

(4) Inversion

79) Match Column I with Column II regarding cross results and select the correct option:

	Column I (Cross / Phenotype)		**Column II (Ratio)**
A.	Monohybrid phenotypic ratio	1.	9 : 3 : 3 : 1
B.	Monohybrid genotypic ratio	2.	1 : 1 : 1 : 1
C.	Dihybrid phenotypic ratio	3.	3 : 1
D.	Dihybrid test cross ratio	4.	1 : 2 : 1

(1) A-4, B-3, C-1, D-2

(2) A-3, B-4, C-1, D-2

(3) A-3, B-1, C-4, D-2

(4) A-1, B-4, C-3, D-2

80) In human beings, haemophilia is an X-linked recessive disorder. If a carrier female ($X^H X^h$) marries a normal male ($X^H Y$), what is the probability that their sons will be haemophilic?

(1) 0%

(2) 25%

(3) 50%

(4) 100%

81) **Statement 1:** Starch synthesis in pea seeds is controlled by a single gene with two alleles (B and b).
Statement 2: Bb heterozygotes produce intermediate-sized starch grains, demonstrating incomplete dominance for starch grain size.

(1) Both Statement 1 and Statement 2 are correct.

(2) Both Statement 1 and Statement 2 are incorrect.

(3) Statement 1 is correct but Statement 2 is incorrect.

(4) Statement 1 is incorrect but Statement 2 is correct.

82) What is the probability of obtaining a plant with genotype AAbbcc from a self-pollinated trihybrid plant (AaBbCc)?

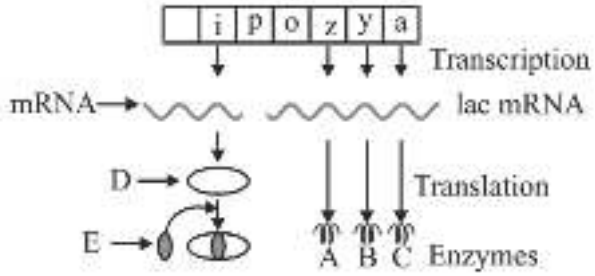
(1) 1/64

(2) 1/16

(3) 1/32

(4) 1/8

83) Given the diagram of the lac operon showing an operon of inducible enzymes. Identify



components and enzymes (A, B, C, D and E)-

(1) A-Beta Galactosidase, B-Permease, C-Transacetylase, D-Repressor protein, E-Inducer (lactose)

(2) A-Beta Galactosidase, B-Permease, C-Transcetylase, D-Inducer (lactose), E-Repressor protein

(3) A-Beta Galactosidase, B-Transacetylase, C-Permease, D-Repressor protein, E-Inducer (lactose)

(4) A-Permease, B-Transacetylase, C-Beta Galactosidase, D-Repressor protein, E-Inducer (Lactose)

84) Identify an initiator - tRNA for translation :-

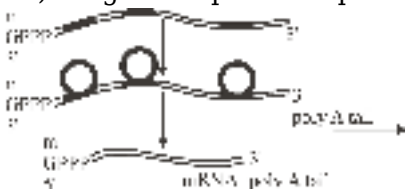
(1) 

(2) 

(3) 

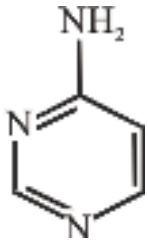
(4) 

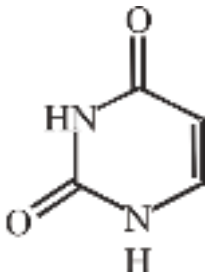
85) Diagram explain the process of :-

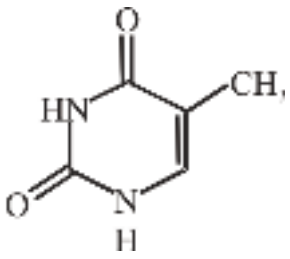


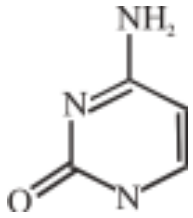
(1) Replication
(2) Translation
(3) Prokaryotic transcription
(4) RNA splicing

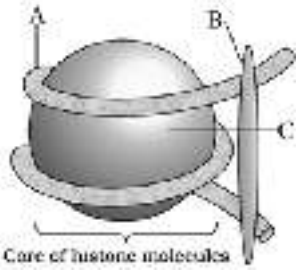
86) Which of the following pyrimidine N-base found in DNA only ?

(1) 

(2) 

(3) 

(4) 



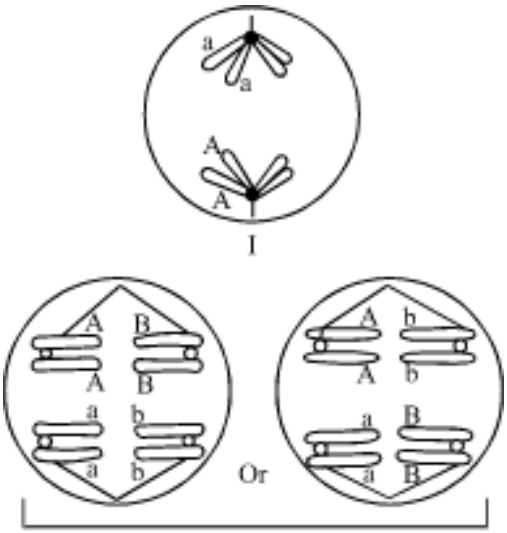
87) In given diagram identify B :

(1) H_1 - Histone
(2) DNA
(3) Histone octamer
(4) Nucleosome



88) Choose the correct statement w.r.t. given figure :

(1) Its genotype is represented as AA + ZW
(2) It produces single type of gametes
(3) It has one more chromosome than female
(4) It is haploid but its female is diploid



89) Two sets of diagrams I and II are given here. Which of the following option is correct?

(1) I – Explains Law of segregation II – Explains Law of dominance
(2) Both I and II explain law of segregation

(3) I – Explains Law of segregation II – Explains Law of Independent assortment

(4) I – Explains Law of segregation II – Explains Law of nondisjunction

90) The Punnett square shown below represents the pattern of inheritance in dihybrid cross, when round (R) is dominant over wrinkled (r) seeds and yellow (Y) is dominant over green (y) seeds.

	RY	Ry	rY	ry
RY	F	J	N	R
Ry	G	K	O	S
rY	H	L	P	T
ry	I	M	Q	U

Find out the odd one w.r.t phenotype –

(1) G

(2) K

(3) H

(4) L

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	3	1	4	2	2	2	1	2	2	2	1	1	2	3	3	3	4	1	1	2
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	4	2	4	3	4	2	1	1	2	3	1	4	2	3	2	3	4	4	4	4
Q.	41	42	43	44	45															
A.	3	3	2	1	2															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	4	2	3	2	3	4	2	3	4	2	3	1	2	2	2	4	3	2	3	4
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	2	2	2	3	3	4	2	2	1	4	2	1	2	2	2	1	2	3	3	2
Q.	86	87	88	89	90															
A.	2	4	3	2	3															

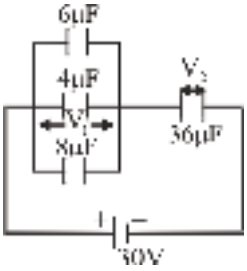
BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	1	2	1	2	4	3	4	2	2	2	3	2	4	3	3	3	2	3	1	1
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	4	3	4	4	1	1	2	1	2	3	2	3	3	1	2	1	2	3	1	4
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	1	3	4	2	1	2	4	1	3	3	1	1	2	4	1	1	2	1	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	4	2	4	2	4	1	1	2	3	2	1	2	4	2	1	1	4	1	2	3
Q.	171	172	173	174	175	176	177	178	179	180										
A.	1	1	1	2	4	3	1	2	3	2										

SOLUTIONS

PHYSICS

1) **Explaining:** To find charge and potential difference across a given capacitor
Concept: This question is based on circuit analysis
Solution:



$V_1 + V_2 = 30V$ and $\frac{V_1}{V_2} = \frac{36}{18} = 2$
Hence $V_1 = 20V$ and $V_2 = 10V$ Charge on 8μF capacitor = $8\mu \times 20 = 160\mu C$
Final Answer: (3)

2)

Explain Question : We need to find force exerted on the wale by the body, from given data.

Concept : This question is based on average force.

Solution :

$$|\vec{F}_{avg}| = \frac{|\Delta \vec{P}|}{\Delta t}$$
$$F_{avg.} = \frac{2mV\sin\theta}{t}$$
$$\Rightarrow F_{avg} = 150\sqrt{3}\ N$$

Final Answer : option (1)

3) **Explaining:** To find energy stored in a capacitor
Concept: This question is based on relation between electric field and energy stored in capacitor
Solution:

$$U = \frac{1}{2}CV^2$$
$$U = \frac{1}{2}\left(\frac{A\in_0}{d}\right)(Ed)^2 = \frac{1}{2}A\in_0 E^2 d$$
Final Answer: (4)

4)

QUESTION EXPLANATION

Consider the three vectors from the original problem: $\vec{a} = -3\hat{i} + 2\hat{j} - \hat{k}$, $\vec{b} = \hat{i} - 3\hat{j} - 5\hat{k}$ and $\vec{c} = -2\hat{i} + \hat{j} - 4\hat{k}$. As established in the original problem, these three vectors can form a closed triangle. The new objective is to find the area of the triangle formed by these three vectors.

CONCEPT

If three vectors form a closed triangle (i.e., their vector sum is zero), the area of that triangle can be found using the cross product of any two of the vectors. The area of a triangle with two vectors, $\vec{a}$ and $\vec{b}$, as its adjacent sides is given by half the magnitude of their cross product. The formula is: $\text{Area} = \frac{1}{2}|\vec{a} \times \vec{b}|$

SOLUTION 1.

First, we need to calculate the cross product of two of the vectors, for instance,

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -3 & 2 & -1 \\ 1 & -3 & 5 \end{vmatrix}.$$

2. Evaluating the determinant:

$\vec{a} \times \vec{b} = \hat{i}((2)(5) - (-1)(-3)) - \hat{j}((-3)(5) - (-1)(1)) + \hat{k}((-3)(-3) - (2)(1))$

$\vec{a} \times \vec{b} = \hat{i}(10 - 3) - \hat{j}(-15 + 1) + \hat{k}(9 - 2)\vec{a} \times \vec{b} = 7\hat{i} + 14\hat{j} + 7\hat{k}$

3. Next, find the magnitude of this resultant cross-product vector:

$\vec{a} \times \vec{b} = \sqrt{7^2 + 14^2 + 7^2} = \sqrt{49 + 196 + 49} = \sqrt{294}$

4. The value $\sqrt{294}$ can be simplified as $\sqrt{49 \times 6} = 7\sqrt{6}$.

5. Finally, calculate the area of the triangle using the formula: $\text{Area} = \frac{1}{2}|\vec{a} \times \vec{b}| = \frac{7\sqrt{6}}{2}$ square units.

CORRECT ANSWER

The correct option is (2).

5) **Explaining:** To find absolute error in time period and percentage error in g

Concept: This question is based on calculation of errors

Solution:

$$\text{Time period } T = \frac{\text{time}}{\text{No. of oscillations}}$$

$$T = \frac{40}{20} = 2 \text{ sec}$$

$$\frac{\Delta T}{T} = \frac{\Delta t}{t} = \frac{1}{40}$$

$$\Delta T = \frac{2}{40} = 0.05 \text{ sec}$$

$$T = 2\pi\sqrt{\frac{\ell}{g}}$$

$$\frac{\Delta g}{g} = \frac{\Delta \ell}{\ell} + \frac{2\Delta T}{T}$$

$\Delta \ell = 0 \quad \Delta T = 0.05$ sec

$\frac{\Delta g}{g} \times 100 = \frac{(2)\,(.05)}{2} \times 100 = 5\%$

Final Answer: (2)

6) **Explaining:** To find dimensional formula of K

Concept: This question is based on dimensional analysis

Solution:

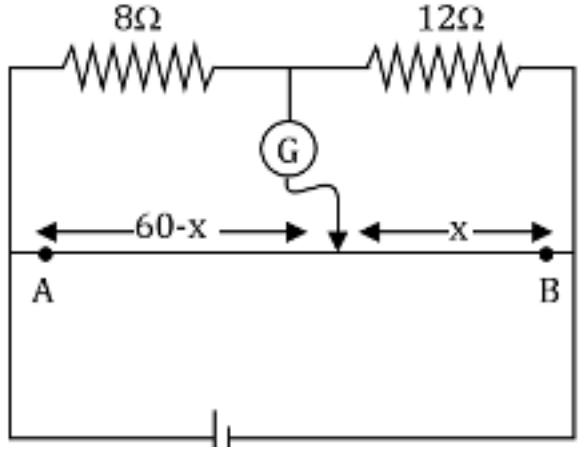
$K = \frac{d^2x}{dt^2} = \frac{[L]}{[T^2]} = [L^1T^{-2}]$

Final Answer: (2)

7) **Explaining:** To find null point on a meter bridge

Concept: This question is based on meter bridge (null point)

Solution:



8 (x) = 12(60 – x)

2x = 180 – 3x ⇒ 5x = 180 ⇒ x = 36 cm

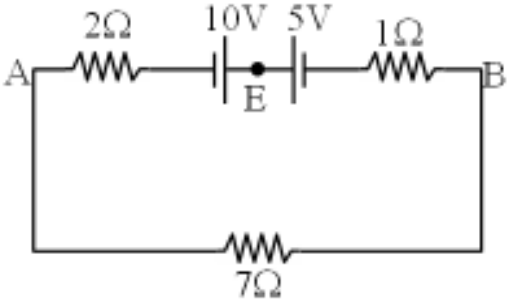
Final Answer: (1)

8) **Explaining:** To find current, potential drop and power in a circuit

Concept: This question is based on potential drop and power

Solution:

$i = \frac{10-5}{10} = \frac{5}{10}A$ = 0.5 A from A to B through E.



$V_A - V_E = -9V$

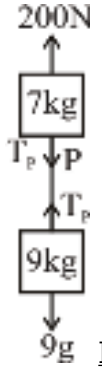
$V_E - V_B = 5.5V$

Power supplied by 10V source = 10X0.5 = 5W

Final Answer: (2)

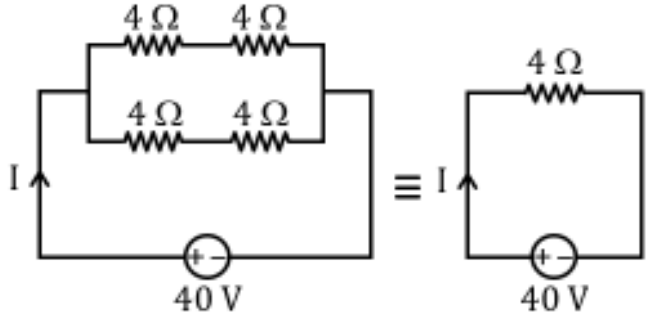
9) **Explaining:** To find current in a circuit
Concept: This question is based on KCL and potential drop
Solution:
IA = 2A
$V_{PR} = 2A \times 4\Omega = 8V$
$I_2 = \frac{8}{2} = 4A$
$I_3 = I_1 + I_2 = 6A$
Final Answer: (2)

10) **Explaining :** Given : Two blocks of masses 7 kg and 5 kg, mass of rope = 4 kg force applied is 200 N.
Ask : Tension at the top of heavy rope.
Concept : This question is based on Motion connected bodies involving
N - T (without pulley or spring)
Solution : $a = \frac{200-160}{16}$



$a = \frac{40}{16} = \frac{5}{2} m/s^2$ **<u>For 9 kg block</u>** $T_P - 9g = 9a$
$T_P = 9g + 9a$
$= 90 + 9 \times \frac{5}{2}$
$= \frac{225}{2} = 112.5N$ **Final Answer : (2)**

11) **Explaining:** To find current in the circuit
Concept: This question is based on wheat stone bridge and KVL
Solution:
It follow Wheatstone bridge so, the equivalent circuit will be

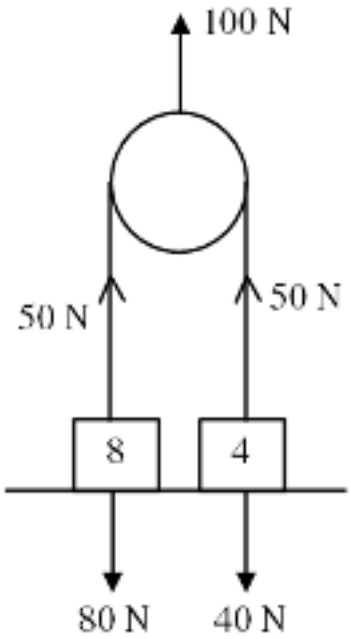


Apply KVL law
–40 + 4I = 0 ⇒ I = 10A

Final Answer: (1)

12) **Explaining :** Match laws/properties to descriptions.
Concept : This question is based on Kirchhoff's first Law = charge conservation; Second Law= energy conservation; resistance depends on physical dimension; resistivity depends on temperature.
Solution :
a–q, b–r, c–p,s, d–s.
Final Answer : (1)

13) **Explaining :** Given : Two blocks are connected by a string, force F 100 N applied on massless pulley.
Ask : Accelerations of blocks.
Concept : This question is based on Pulley block system.
Solution :



As F = 100 N,
Tension in string is 50 N
So m_2 can not move, its acceleration is zero, for m_1
$50 - 4g = 4a \Rightarrow a = 2.5\ m/s^2$
Final Answer : (2)

14) **Question Explanation:** Asking about the maximum applied voltage across the series combination of C_1 and C_2 for the given breakdown voltage.

Concept: Breakdown strength of Capacitors

Solution:
We know Q = CV

Hence $(Q_1)_{max}$ = 6 mC while $(Q_2)_{max}$ = 12mC

However in series charge is same so maximum charge on C_2 will also be 6mC (and not 12mC) and hence potential difference across C_2 will be

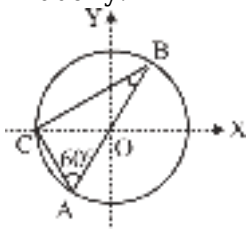
$$V_2 = \frac{6mC}{3\mu F} = 2kV$$ and as in series

$V = V_1 + V_2$

So V_{max} = 6kV + 2kV = 8kV

Final answer: option (3)

15) **Explaining:** To find electric field, potential energy, net force and potential at a given point
Concept: This question is based on multiple concepts of energy, force and potential
Solution:
Electric field at point O is vector sum of electric field due to all three point charges individually.



$$\vec{E}_O = \vec{E}_A + \vec{E}_B + \vec{E}_C$$

and $\vec{E}_A = -\vec{E}_B$

So, $\vec{E}_O = \vec{E}_C$

$$= \frac{2q/3}{4\pi\varepsilon_0 R^2} \text{along } \vec{OC} = \frac{q}{6\pi\varepsilon_0^2} \text{along } \vec{OC}$$

$$= -\frac{q}{6\pi\varepsilon_0 R^2}\hat{i}$$

From geometry, we can find that ∠ABC = 30° and ∠ACB = 90°

So AB = 2R; AC = R; BC = $\sqrt{3}R$
Total potential energy of the system is,

$$U = \frac{1}{4\pi\varepsilon_0}\left[\frac{q/3 \times q/3}{2R} - \frac{q/3 \times 2q/3}{R} - \frac{q/3 \times 2q/3}{\sqrt{3}R}\right] \neq 0$$

Force of interaction between B and C is,

$$F = \frac{q/3 \times 2q/3}{4\pi\varepsilon_0 \times (\sqrt{3}R)^2} = \frac{q^2}{54\pi\varepsilon_0 R^2}$$

Potential at point O is , $$V = \frac{q/3 + q/3 - 2q/3}{4\pi\varepsilon_0 R} = 0$$

Final Answer: (3)

16) **Explaining:** To find acceleration and velocity of a moving particle
Concept: This question is based on kinematics of motion
Solution:
$x = 3t^2 - 6t \qquad y = t^2 - 2t$

$v_x = \frac{dx}{dt} = 6t - 6 \quad v_y = \frac{dy}{dt} = 2t - 2$

at t = 1 sec → $v_x = 0$ and $v_y = 0$

hence $v = \sqrt{v_x^2 + v_y^2} = 0$

Final Answer: (3)

17) **Explaining:** To find the separation between the trains after both have stopped
Concept: This question is based on relative motion
Solution:
Final separation
= Initial separation – |Relative displacement|
= 300 – [(1/2) × 40 × 10 + (1/2) × 20 × 8]
= 20 m
Final Answer: (4)

18) **Explanation :** Find angle between vector
Concept : This question is based on $R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$
Solution :

$\vec{A} = \vec{B} + \vec{C}$

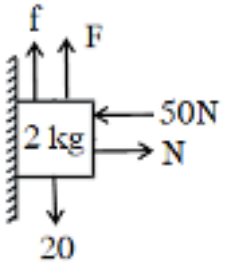
$\vec{B} = \vec{A} - \vec{C}$

$B^2 = A^2 + C^2 - 2AC\cos\theta.$

$\cos\theta = \frac{3}{5}, \qquad \theta = \cos^{-1}\left(\frac{3}{5}\right)$

Final Answer : (1)

19) **Explain the Question :** The block shown in the figure in so heavy, that the friction alone can not hold it. We need to find the minimum upward force F which will ensure equilibrium of the block.
Concept : This question is based on block against rough vertical wall.



Solution : $f_{max} = \mu N = 0.2 \times 50 = 10$ N

F + f = 20

F + 10 = 20 ⇒ F = 10 N

Final Answer : 1

20)

QUESTION EXPLANATION
The objective is to find the value of the expression (cotθ + cscθ).

CONCEPT
This requires finding the values of other trigonometric functions based on the given sine value and the quadrant.

SOLUTION
We are given $\sin\theta = \frac{12}{13}.$
cosθ :
$\cos^2\theta = 1 - \sin^2\theta = 1 - \left(\frac{12}{13}\right)^2 = 1 - \frac{144}{169} = \frac{169-144}{169} = \frac{25}{169}$.
$\cos\theta = \pm\sqrt{\frac{25}{169}} = \pm\frac{5}{13}.$
Since `θ` is in the second quadrant, cosθ must be negative. Therefore, $\cos\theta = -\frac{5}{13}.$
Calculate the required expression:
$\cos\theta + \csc\theta = -\frac{5}{12} + \frac{13}{12} = \frac{13-5}{12} = \frac{8}{12} = \frac{2}{3}.$

CORRECT ANSWER
The correct option is (2).

21) **Explaining:** To find point at which electric field intensity is zero
Concept: This question is based on electric field
Solution:
For E = 0, $E_1 = E_2$
Let x is the distance from –Q on left side
$\frac{KQ}{x^2} = \frac{2KQ}{(x+3)^2}$
$\Rightarrow \sqrt{2}x = x + 3 \Rightarrow x = \frac{3}{\left(\sqrt{2}-1\right)} = 3\left(\sqrt{2}+1\right) = \left(3\sqrt{2}+3\right)$
from 2Q
$3 + x = \left(3\sqrt{2}+3\right) + 3 = \left(3\sqrt{2}+6\right)$
Final Answer: (4)

22) **Question Explanation:** Calculating impulse magnitude from a position-time graph.
Velocity changes from 1m / s to - m / s

Concept: Impulse as change in momentum.

Solution:
Velocity (v) = slope of x – t graph

$V = \frac{\Delta x}{\Delta y} = \frac{2}{2} = 1 m/s$

Velocity changes from + 1m/s to –1 m/s at peaks

Magnitude of change in velocity (Δv) = |(–1) –1|= 2m/s

Impulse (I) = m × Δv

I = 0.4kg × 2m/s

I = 0.8 kg-m/sec

Final Answer: Option (2)

23) **Explaining :** given : volume = 9 m^3, H = 10 m, power input = 10 kw, t = 5 min, s = 10^3 kg/m^3

ask : efficiency of the pump

Concept : This question is based on power

Solution :

$$\eta = \left(\frac{P_{out}}{P_{input}}\right) \times 100 = \frac{\frac{mgh}{t}}{10 \times 10^3}$$

$$= \frac{10^3 \times 9 \times 10 \times 10}{5 \times 60 \times 10 \times 10^3} \times 100 = \frac{900}{30} = 30\%$$

Final Answer : (4)

24) **Explaining:** To find electric field due to uniformly charge sphere

Concept: This question is based on electric field

Solution:

$$E_1 = \frac{kQ}{R^2}$$

$$E_2 = \frac{kQ}{R^2}$$

$$E_3 = \frac{kQ \cdot R}{(2R)^3} = \frac{kQ}{8R^2} = \frac{E_1}{8}$$

Thus, $E_1 = E_2 > E_3$

Final Answer: (3)

25) **Explaining:** To find the time when to body meet

Concept: This question is based on kinematics of motion (motion in the gravity)

Solution:

$$\because S = Ut + \frac{1}{2}at^2$$

$$\therefore h = 98t - \frac{1}{2} \times 10 \times t^2 \quad ...(1)$$

$$\& \; h = 98(t-4) - \frac{1}{2} \times g \times (t-4)^2 \quad ...(2)$$

From equation (1) & (2)

$$98t - \frac{1}{2} \times g \times t^2 = 98t - 392 - \frac{1}{2} \times gt^2 - 78.4 + 39.2t$$

39.2t = 470.4

t = 12 sec.

Final Answer: (4)

26)

Explain Question : We need to select the correct SI unit for potential gradient.

Concept : Dimensions of physical quantities and SI units.

Solution :

Potential gradient $\Rightarrow \frac{\text{Potential}}{\text{length}} = \frac{V}{m}$

Final Answer : option (2)

27) **Explaining :** To find the magnitude of force when charges are placed at corners of cube of side length b.

Concept : This question is based on coulombs law

Solution :

$F \propto \frac{1}{a^2} \quad F_1 \propto \frac{1}{b^2} \Rightarrow F_1 = \frac{Fa^2}{b^2}$

Final Answer : (1)

28) **Explaining :** Energy dissipated when 100 μF capacitor charged to 200 V is discharged through resistor.

Concept : This question is based on All stored energy converts to heat.

Solution :

C = 100μf V = 200V

Heat generated = $\frac{1}{2} CV^2$

$\frac{1}{2} \times 100 \times 10^{-6} \times 200 \times 200 = 2J$

Final Answer : (1)

29)

Explain the Question :- To find the mobility of electrons provided the drift velocity = $5 \times 10^6 ms^{-1}$

Concept :-This question is based on drift velocity

Solution :-

L = 40 cm = 0.4 m

v = 10 V

$v_d = 5 \times 10^{-6}\ ms^{-1}$

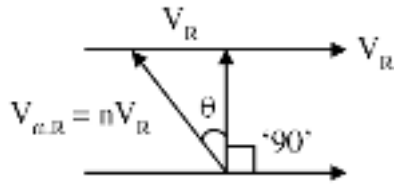
$\mu = \frac{v_d}{E} = \frac{v_d L}{V} = \frac{5 \times 10^{-6} \times 0.4}{10}$

$= 2 \times 10^{-7}\ m^2 v^{-1} s^{-1}$

Final answer :- (2)

30)

Explanation the Question:

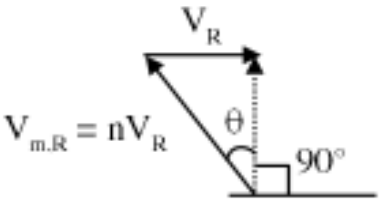


At what angle to the stream direction must be boat move to cross the river along shortest path?

Concept:

This question is based on To cross the river in shortest possible path man/boat should swim at obtuse angle [90 + θ] with the river/stream flow.

Solution:



from the diagram, $\sin\theta = \frac{V_R}{V_{m,R}}$

$\sin\theta = \frac{V_R}{nV_R} \Rightarrow \sin\theta = \frac{1}{n}$

$\theta = \sin^{-1}\left(\frac{1}{n}\right)$

The boat show move at on angle of $90 + \sin^{-1}\left(\frac{1}{n}\right)$ or $\frac{\pi}{2} + \sin^{-1}\left(\frac{1}{n}\right)$ with the river flow

Final Answer: (3)

31)

Concept:

A vector in the direction of $\vec{A}$ with magnitude equal to $|\vec{B}|$ is given by :

$\vec{V} = |\vec{B}|\hat{A}$

where $\hat{A}$ is the unit vector of $\vec{A}$.

Formula:

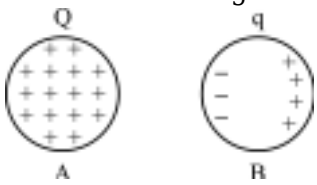
1. Magnitude of $\vec{A}$: $|\vec{A}| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$
2. Magnitude of $\vec{B}$: $|\vec{B}| = \sqrt{7^2 + 24^2} = \sqrt{49 + 576} = \sqrt{625} = 25$
3. Unit vector in direction of $\vec{A}$: $\hat{A} = \frac{\vec{A}}{|\vec{A}|} = \frac{3\hat{i} + 4\hat{j}}{5} = \frac{3}{5}\hat{i} + \frac{4}{5}\hat{j}$
4. Required vector: $\vec{V} = |\vec{B}|\hat{A} = 25\left(\frac{3}{5}\hat{i} + \frac{4}{5}\hat{j}\right)$

$\vec{V} = 15\hat{i} + 20\hat{j}$

Final Answer:

$\vec{V} = 15\hat{i} + 20\hat{j}$

32) **Explaining :** To determine the nature of force between similarly charged bodies.
Concept : This question is based on Coulomb's law and induction.
Solution :
Two similar charged bodies may attract each other only in special case.



If Q and q are similar charge and induced charge of body B is at smaller distance to A as compare to positive charge of B. If q < Q there there is a possibility that attraction is more than repulsion.
Final Answer : (4)

33) **Explaining:** To find equation of trajectory of the particle
Concept: This question is based on kinematics of motion in 2 dimensions
Solution:
$u_x = u_0$, $x = u_0 t$, $t = x/u_0$
$u_y = (a\omega \cos\omega t)j$, $y = a \sin\omega t$
$y = a\sin\left(\frac{\omega x}{u_0}\right)$
Final Answer: (2)

34)

QUESTION EXPLANATION

The question provides the equation of a curve, $y = \frac{1}{4\sqrt{3}}x^2$.
It asks for the coordinates of a point P on this curve where the tangent to the curve makes an angle of 30° with the positive x-axis.

CONCEPT
The slope of the tangent to a curve at any point is given by the first derivative of the curve's equation at that point, $m = \frac{dy}{dx}$.
The slope is also related to the angle `θ` it makes with the x-axis by m = tanθ.
We can set the derivative equal to tan(30°) to find the x-coordinate of point P, and then use the curve's equation to find the corresponding y-coordinate.

SOLUTION
1. First, find the derivative of the curve's equation to get the slope `m`:
$m = \frac{dy}{dx} = \frac{d}{dx}\left(\frac{1}{4\sqrt{3}}x^2\right) = \frac{1}{4\sqrt{3}}(2x) = \frac{x}{2\sqrt{3}}$.
2. The slope is also given by the tangent of the angle:
$m = \tan(30^\circ) = \frac{1}{\sqrt{3}}$.
3. Set the two expressions for the slope equal to each other to find the x-coordinate of P:

$\frac{x}{2\sqrt{3}} = \frac{1}{\sqrt{3}}$.

4. Solving for x gives x = 2.

5. Now, substitute x = 2 back into the original equation of the curve to find the y-coordinate of P:

$y = \frac{1}{4\sqrt{3}}(2)^2 = \frac{4}{4\sqrt{3}} = \frac{1}{\sqrt{3}}$.

6. Therefore, the coordinates of point P are $\left(2, \frac{1}{\sqrt{3}}\right)$.

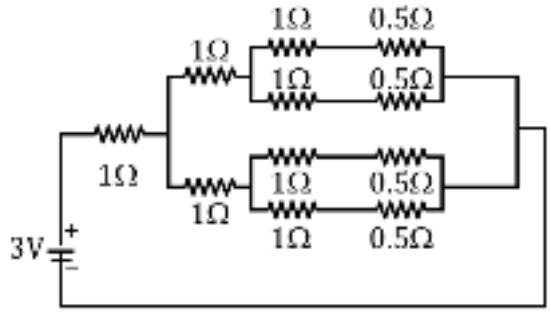
CORRECT ANSWER

The correct option is (3).

35) **Explaining:** To find current in a circuit

Concept: This question is based on circuit analysis

Solution:



From above figure, $R_{eq} = \frac{15}{8}\Omega$

So, $I = \frac{V}{R} = \frac{3}{\frac{15}{8}} = \frac{24}{15}A = \frac{8}{5}A$

Final Answer: (2)

36) **Explaining :** given : U - x and F - x curve are given, find correct graph

Concept : This question is based on Equilibrium, $\vec{F} = -\Delta U$

Solution :

$F = -\frac{dU}{dx}$

Potential energy U is decreasing with x,

so, $\frac{du}{dx} < 0$, hence F > 0 and vice - versa

Final Answer : (3)

37) **Explaining :** If K.E. increases by 4%, then momentum will increase

Concept : This question is based on Relation between K.E. and momentum

Solution :

$K \propto p^2 \Rightarrow \frac{\Delta K}{K} = \frac{2\Delta p}{p}$

So $\frac{\Delta p}{p} = \frac{1}{2}\frac{\Delta K}{K}$ = 2%

Final Answer : (4)

38) **Explaining:** To find condition for equilibrium of q
Concept: This question is based on net force (coulomb's law) and equilibrium conditions
Solution:
if q is + ve
axial displacement along AB → stable equilibrium equational displacement along CD → unstable equilibrium
if q is – ve
axial displacement along AB → unstable equilibrium equational displacement along CD → stable equilibrium
Hence all of the above
Final Answer: (4)

39) **Explaining:** To find ratio of tension in strings
Concept: This question is based on tension in the strings
Solution:

$$T_1 = (12+3)\,a \text{ and } T_2 = 3a$$

$$\therefore \quad \frac{T_1}{T_2} = \frac{15a}{3a} = \frac{5}{1}.$$

Final Answer: (4)

40) **Explaining:** To find potential difference V across the capacitance when the temperature of the block rises by ΔT
Concept: This question is based on energy of a capacitor
Solution:

$$\frac{1}{2}CV^2 = ms\Delta T \Rightarrow V = \sqrt{\frac{2ms\Delta T}{C}}$$

Final Answer: (4)

41) **Explaining:** To find acceleration of the particle at a given position
Concept: This question is based on a relation between x, v and a
Solution:
$x = \sqrt{V+1}$
$x^2 = V + 1$
$V = x^2 - 1$
$a = V\frac{dV}{dx} = (x^2 - 1)(2x)$
at x = 5 a = 24 × 10 = 240 unit
Final Answer: (3)

42) **Explaining :** given : k = 100 N/m, x_1 = 10 cm, x_2 = 20 cm,
ask : work done by external agent in stretching a spring
Concept : This question is based on work done by spring
Solution :

$U_2 = \frac{1}{2}kx_2^2$

$U_1 = \frac{1}{2}kx_1^2$ $W_{ext} = \Delta U = U_2 - U_1$

$= \frac{1}{2}kx_2^2 - \frac{1}{2}kx_1^2 = \frac{1}{2}k(x_2^2 - x_1^2)$

$= \frac{1}{2} \times 10000 \times \left(\left(\frac{20}{100}\right)^2 - \left(\frac{10}{100}\right)^2\right) = 150J$

Final Answer : (3)

43) **Question Explanation:** Two identical charged balls repel and hang symmetrically. Find tension using equilibrium.

Concept:
For each ball:

A. Horizontal: Coulomb force
B. Vertical: Weight
C. Tension balances both components

Formula: $F = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{a^2}$ $T \sin\theta = F$, $T \cos\theta = mg$

Geometry: $\sin\theta = \frac{a/2}{\ell} = \frac{a}{2\ell}$

Calculation

From horizontal balance: $T \sin\theta = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{a^2}$ Substitute $\sin\theta = \frac{a}{2\ell}$: $T \sin\theta = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{a^2}$

$T \cdot \frac{a}{2\ell} = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{a^2}$

$T = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{a^2} \cdot \frac{2\ell}{a}$

$T = \frac{q^2\ell}{2\pi\varepsilon_0 a^3}$

Final Answer: Option (2)

44) **Explaining:** To find volume charge density of a spair
Concept: This question is based on gauss's law
Solution:
Consider a spherical symmetric surface
$\phi = \phi_0 r^4$
$Q = \epsilon_0\phi_0 r^4$ [From Gauss' Theorem]
$\rho = \frac{dQ}{dV} = \frac{dQ}{4\pi r^2 dr} = \frac{\epsilon_0\phi_0 4r^3}{4\pi r^2} \Rightarrow \rho = \frac{\epsilon_0\phi_0 r}{\pi}$
Final Answer: (1)

45) **Explaining :** To find power generated by a constant force F applied on a body
Concept : This question is based on power
Solution :

$$F = \frac{mdv}{dx}$$

$$\therefore Fdt = mdv \Rightarrow v = \frac{F}{m}t$$

Now $P = f \times v = F \times \frac{F}{m}t = \frac{F^2t}{m}$

If force and mass are constants then $P \propto t$.

Final Answer : (2)

CHEMISTRY

46) **Explaining :** If 0.01 M solution of an electrolyte has a resistance of 40 ohms in a cell having a cell constant of 0.4 cm^{-1}, then its molar conductance in ohm^{-1} cm^2 mol^{-1} is :

Concept : This question is based on molar conductance.

Solution :

Molarity = 0.01 M ; Resistance = 40 ohm;

Cell constant $\frac{l}{A} = 0.4\ cm^{-1}$.

Specific conductivity (κ) $= \frac{\text{cell constant}}{\text{resistance}} = \frac{0.4}{40} = 0.01\ ohm^{-1}\ cm^{-1}$

Molar conductance $(\wedge_m) = \frac{1000\kappa}{\text{Molarity}}$

$= \frac{1000 \times 0.01}{.01} = 10^3\ ohm^{-1}\ cm^2\ mol^{-1}$

Final Answer : (4)

47) **Explaining :** Calculate Λ^{∞}_{HOAc} using appropriate molar conductances of the electrolytes listed above at infinite dilution in H_2O at 25°C :

Concept : This question is based on molar conductances.

Solution :

$\Lambda^{\infty}_{HCl} = 426.2$ ...(i)

$\Lambda^{\infty}_{AcONa} = 91.0$ ...(ii)

$\Lambda^{\infty}_{NaCl} = 126.5$ ...(iii)

Λ^{∞}_{AcOH} = (i) + (ii) – (iii) = [426.2 + 91.0 – 126.5] = 390.7 **Final Answer :** (2)

48) **Explaining :** How much chlorine will be liberated on passing one ampere current for 30 min. through NaCl solution?

Concept : This question is based on Faraday's law.

Solution :

Charge = 1 × 30 × 60 = 1800 coulomb

96500 liberates 35.5 gm of Cl

1800 coulomb liberates

$\frac{35.5 \times 1800}{96500} = 0.66\ gm\ of\ Cl$

Final Answer : (3)

49) **Explaining :** For the electrochemical cell, M | $[M^+$ || X^- | X, $E°(M^+/M) = 0.44$ V and E°

(X/X^-) = 0.33 V.
From this data one can deduce that:
Concept : This question is based on ECS.
Solution :
For the given cell M | M^+ || X^+ | X, the reaction can be given by
R.H.S. : Reduction $X + e^- \rightarrow X^-$; E^0 = 0.33 V
L.H.S. : Oxidation $M \rightarrow M^+ + e^-$; E^0 = 0.49 V
Adding both half cell reactions, we get $M + X \rightarrow M^+ + X$
$$E^o_{cell} = E^o_{RHS} = E^o_{LHS} = 0.33 - 0.44 = -0.11\,V$$
Since E_{Cell} = –ve, the reaction can not be spontaneous in forward direction but can proceed spontaneously if the direction is reversed.
Final Answer : (2)

50) **Explaining :** Using same quantity of current, which among Na, Mg, and Al is deposited to the highest extent during electrolysis.
Concept : This question is based on electrolysis.
Solution :
According to Faraday's law of electrolysis, the amount of metal deposited on an electrode is directly proportional to the amount of electricity passed and equivalent weight of metal. Now equivalent weight of sodium is 23, magnesium is $\frac{24.3}{2}$ i.e. 12.15 and of aluminium is $\frac{27}{3}$ i.e. 9. So, sodium which has highest equivalent weight among Na, Mg and Al, is deposited to highest extent during electrolysis.
Final Answer : (3)

51) **Explaining :** Aluminium oxide may be electrolysed at 1000°C to furnish aluminium metal (At. Mass = 27 amu; 1 Faraday = 96,500 Coulombs).
The cathode reaction is– $Al^{3+} + 3e^- \rightarrow Al$ To prepare 5.12 kg of aluminium metal by this method we require electricity of :
Concept : This question is based on Faraday's law.
Solution :
1 mole of e^- = 1F = 96500 C
27g of Al is deposited by 3 × 96500 C
5120 g of Al will be deposited by $= \frac{3 \times 96500 \times 5120}{27} = 5.49 \times 10^7 C$
Final Answer : (4)

52) **Explaining :** Which of the following will form a cell with the highest voltage?
Concept : This question is based on Nernst equation.
Solution : Here Ag^+ acts as oxidising agent and Co^{2+} acts as reducing agent. So, cell can be represented as Co^{2+} | Co^{3+}(aq.) || Ag^+(aq.) | Ag Using Nernst equation,
$$E_{cell} = E^*_{cell} - \frac{0.0591}{n}\log\left|\frac{[Co^{3+}(aq)][Ag]}{[Ag^+(aq)][Co^{2+}]}\right|$$ So, lesser the logarithmic term higher will be the value of E_{cell}. Among the given combinations logarithmic term is lesser for 0.1 M Co^{3+} and 2M Ag^+ and hence it will form a cell of highest voltage. **Final Answer :** (2)

53) **Explaining :** What is the E^o_{cell} for the reaction

$Cu^{2+}(aq) + Sn^{2+}(aq) + Cu_5 + Sn^{4+}(aq)$ at 25°C if the equilibrium constant for the reaction is 1×10^6 ?

Concept : This question is based on Nernst equation.

Solution :

$$E^\circ_{cell} = \frac{2.303RT}{nF}\log K_{eq} = \frac{.0591}{n}\log K_{eq}$$

$$-\frac{.0591}{2}\log 10^6 - .0591 \times 3 - 0.1773\ V$$

Final Answer : (3)

54) **Explaining :** In electrolysis of dilute H_2SO_4, what is liberated at anode?

Concept : This question is based on Hydrogen electrode.

Solution :

In electrolysis of dilute H_2SO_4, O_2 is liberated at anode.

It can be summarised as follows :-

$H_2SO_4 \rightleftharpoons 2H^+ + SO_4^{2-}$

$H_2O \rightleftharpoons H^+ + OH^-$

During electrolysis, the hydrogen ions migrate towards the cathode and are collected here in the form of hydrogen gas.

Following reactions take place

At cathode : $2H^+ + 2e^- \longrightarrow H_2$

$4H^+ + 4e^- \longrightarrow 2H_2$

At anode : $2H_2O \longrightarrow O_2 + 4H^+ + 4e^-$

Overall reaction is

$4H^+ + 2H_2O \rightleftharpoons 2H_2 + O_2 + 4H^+$

Thus, we see that hydrogen is produced at cathode and oxygen at anode during electrolysis.

Final Answer : (4)

55) **Explaining :** The chemical reaction, $2AgCl_{(s)} + H_{2(g)} \rightarrow 2HCl_{(aq)} + 2Ag_{(s)}$ taking place in a galvanic cell is represented by the notation.

Concept : This question is based on Galvanic cell.

Solution :

$2AgCl_{(s)} + H_{2(g)} + 2HCl_{(aq)} + 2Ag_{(s)}$

The activities of solids and liquids are taken as unity and at low concentrations, the activity of a solute is approximated to its molarity.

The cell reaction will be $Pt_{(s)} \mid H_{2(g)}$, 1bar $\mid H^+_{(aq)}$1M$\mid AgCl_{(aq)}$ 1M$\mid Ag_{(s)}$

Final Answer : (2)

56) **Explaining :** Specific conductance of 0.1 M HNO_3 is 6.3×10^{-2}. The molar conductance of the solution is :

Concept : This question is based on Specific conductance.

Solution :

Molar conductance of solution is related to specific conductance as follows :

$$\Lambda_m = K \times \frac{1000}{C} \qquad \ldots(1)$$

where C is molar conc^n.

Putting K = 6.3×10^{-2} ohm^{-1} cm^{-1} and C = 0.1M in equ (1), we get

$$\Lambda_m = \left(6.3 \times 10^{-2} \text{ohm}^{-1}\text{cm}^{-1}\right) \times \frac{1000}{(0.1\,\text{mol/cm}^3)}$$

$= 6.3 \times 10^{-2} \times 10^4$ ohm^{-1}cm^2 mol^{-1}

$= 630$ ohm^{-1} cm^2 mol^{-1}

Final Answer : (3)

57) **Explaining :** At 298K the standard free energy of formation of H_2O (□) is –237.20 kJ/mole while that of its ionisation into H^+ ion and hydroxyl ions is 80 kJ/mole, then the emf of the following cell at 298 K will be [Take Faraday constant F = 96500C]

$H_2(g, 1\text{ bar}) \mid H^+(1M) \mid\mid OH^-(1M) \mid O_2(g, 1\text{ bar})$

Concept : This question is based on ΔG° concept.

Solution :

Cell reaction

cathode : $H_2O(\ell) + \frac{1}{2}O_2(g) + 2e^- \rightarrow 2OH^-(aq)$

anode: $H_2(g) \rightarrow 2H^+(aq) + 2e^-$

$$H_2O(\ell) + \frac{1}{2}O_2(g) + H_2(g) \rightarrow 2H^+(aq) + 2OH^-(aq)$$

Also we have

$$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(\ell);\ \Delta G_f^\circ = -237.2 \text{ kJ/mole}$$

$H_2O(l) \rightarrow H^+(aq) + OH^-(aq)$;

$\Delta G^\circ = 80$ kJ/mole

Hence for cell reaction

$\Delta G^\circ = -237.2 + (2 \times 80) = -77.20$ kJ/mole

$$\therefore E^\circ = -\frac{\Delta G^\circ}{nF} = \frac{77200}{2 \times 96500} = 0.40V$$

Final Answer : (1)

58) **Explaining :** Isomerism

Concept : This question is based on isomerism

Solution : Ambidentate ligand → Linkage isomerism

Final Answer : (2)

59) **Explaining :** Type of isomerism

Concept : This question is based on type of isomerism

Solution : NO_2 is ambidentate ligand.

Final Answer : (2)

60) **Explaining :** Geometrical isomers of Ma_3b_3.

Concept : This question is based on isomerism

Solution : Fac-mer isomers.

Final Answer : (2)

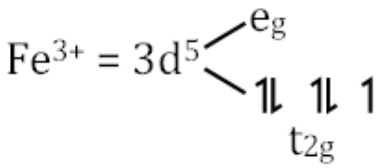
61) **Explaining :** Crystal Field splitting Energy
Concept : This question is based on Crystal Field splitting Energy
Solution : $\Delta_t = \frac{4}{9}\Delta_0$
$[CoCl_4]^{2-} = \frac{4}{9} \times 18000 = 8000\,cm^{-1}$ **Final Answer :** (4)

62) **Explaining :** Conduction
Concept : This question is based on Werner's theory
Solution : Electricity is conducted due to the presence of free ions.
Final Answer : (3)

63) **Explaining :** Type of ligand
Concept : This question is based on type of ligand.
Solution : EDTA have 2 'N' and 4 'O' donar atoms.
Final Answer : (2)

64) **Explaining :** Geometry of complex
Concept : This question is based on crystal field theory
Solution :
$[Ni(CN)_4]^{2-}$ is square planar and other three are tetrahedral.
Final Answer : (3)

65) **Explaining :** magnetic character
Concept : This question is based on crystal field theory
Solution :



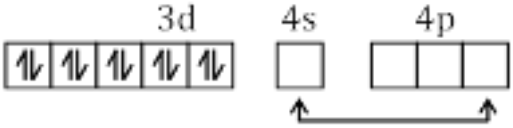
Final Answer : (4)

66) **Explain Question:**
Identify the colorless complex.
Concept:
CFT and its application.
Solution:
Color is shown due to d-d transition in complex.
d-d transition not possible for d° and d° configuration and they are generally colorless.
(1) $[Cr(H_2O)_6]Cl_3 \rightarrow Cr^{+3}\ \ 3d^3$ - color possible
(2) $[Sc(H_2O)_6]Cl_3 \rightarrow Sc^{+3}\ \ 3d^0$ - colorless
(3) $[Ti(H_2O)_6]^{+3} \rightarrow Ti^{+3}\ \ 3d^1$ - color possible
(4) $KMnO_4$ - Mn^{+7} – $3d^0$, but purple , it is due to
Legend- to- metal change transfer, transition.
Final Answer: Option (2)

67) **Explaining :** Application of CFT
Concept : This question is based on application of CFT
Solution : $[Fe(CN)_6]^{3-}$ is most stable due to CFSE.
Final Answer : (2)

68) **Explaining :** CFT
Concept : This question is based on CFT
Solution :
$[Co(NH_3)_6]^{3+}$
Six pairs of electrons, one from each NH_3 molecule, occupy the six hybrid orbitals. Thus, the complex has octahedral geometry and is diamagnetic because of the absence of unpaired electron. In the formation of this complex, since the inner d orbital (3d) is used in hybridisation, the complex, $[Co(NH_3)_6]^{3+}$ is called an inner orbital or low spin or spin paired complex.
Final Answer : (2)

69) **Explaining :** CFT
Concept : This question is based on CFT
Solution :
$[Co(CO)_4]^-$ oxidation state of central metal ion = x + 0 = –1
x = –1
So configuration is $3d^8\ 4s^2$ pairing will be there as CO is SFL



sp^3, tetrahedral
Final Answer : (3)

70)

Explain question: Identify correct IUPAC name of given cobalt coordination complex. oxidation state, and name the counter-ion).
Concept: Coordination nomenclature
Solution: Lignad names (alphabetically, ignoring multiplicative prefixes): ammine, chlorido, nitrito-O (ONO binds through O), pyridine – use bis(pyridine) for two pyridine ligands. Metal oxidation state = +3. Counter-ion = nitrate.
Assembled IUPAC name (alphabetical order):
Diamminechloridonitrito-O-dipyridine cobalt (III) nitrate.
Final Answer: Option (3)

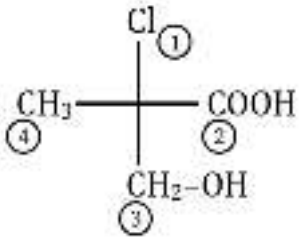
71) **Explaining :** Types of ligand
Concept : This question is based on types of ligand
Solution :
When a ligand is bound to a metal ion through a single donor atom, as with Cl^-, H_2O or NH_3, the ligand is said to be unidentate.
When a ligand can bind through two donor atoms as in $H_2NCH_2CH_2NH_2$ (ethane-1,2-diamine) or $C_2O_4^{2-}$ (oxalate), the ligand is said to be bidentate and when several donor atoms are present

in a single ligand as in $N(CH_2CH_2NH_2)_3$, the ligand is said to be polydentate. $EDTA^{4-}$ is a polydentate ligand.
Final Answer : (4)

72) **Explaining :** Relation between two compound.
Concept : This question is based on POS and mirror image.
Solution :
Meso means optically inactive & d means optically active so both are not mirror images of each other.
Final Answer : (2)

73)

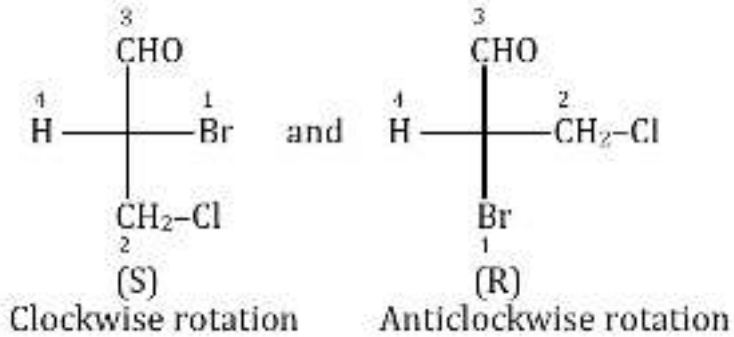
Explaining : Configuration of chiral-carbon.
Concept : This question is based on R and S configuration.
Solution :



S-configuration
Final Answer : (2)

74)

Explaining : Relation between compound.
Concept : This question is based on R and S configuration.
Solution :



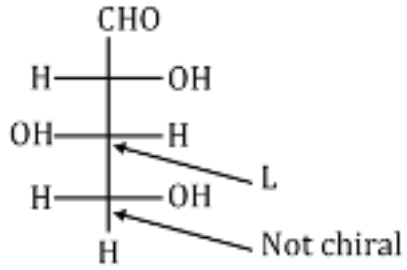
Final Answer : (1)

75) No P.O.S. , No C.O.S. (Asymmetric molecule)

76)

Explaining : Which can show L-Configuration.
Concept : This question is based on (D) and (L) Configuration.
Solution :

See configuration of last chiral carbon



Final Answer : (2)

77)

Explaining : Correct statement about (A).
Concept : This question is based on Not mirror image.
Solution : 'A' and 'B' are diastereomers.
Final Answer : (1)

78) **Explaining :** A/R based question.
Concept : This question is based on Elimination.
Solution :

$$CH_3—\underset{\underset{CH_3}{|}}{CH}—CH_2CH_3 \xrightarrow{\text{Alc. KOH}} \underset{\text{(Major)}}{\underset{\text{(Trans product)}}{CH_3—CH=CH_2CH_3}}$$

Stability = $\underset{\text{2-Butene}}{H_3C—CH=CH—CH_3} > \underset{\text{(1-Butene)}}{H_2C = CH—CH_2—CH_3}$

Final Answer : (2)

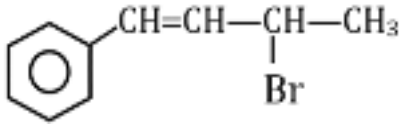
79) **Explaining :** Statement based question.
Concept : This question is based on Reaction of RX with AgCN and $NaNO_2$
Solution :
$RX + AgCN \longrightarrow R—NC$
$RX + NaNO_2 \longrightarrow R—O—N = O$
Final Answer : (2)

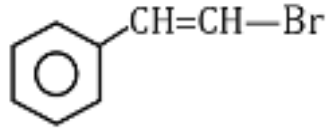
80) **Explaining :** A,R based question.
Concept : This question is based on Grignard reagent.
Solution :

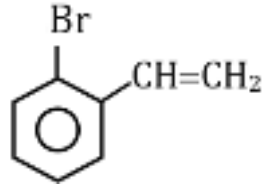
$$RX + \xrightarrow{\text{Na/dry ether}} R-R$$
$$R-X \xrightarrow[\text{Mg/dry ether}]{} RMgX$$

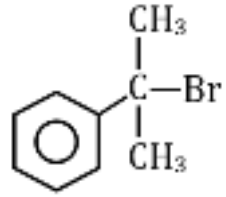
Final Answer : (2)

81) **Explaining :** Match the column-
Concept : This question is based on Type of Halide.
Solution :

(A)  → (R) Allylic halide

(B)  → (S) Vinylic halide

(C)  → (P) Aryl halide

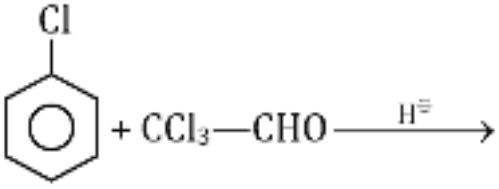
(D)  → (Q) Benzylic halide

Final Answer : (1)

82) **Explaining :** Match the column-

Concept : This question is based on use of Haloalkane and product name based.

Solution :

(A)  (Q) DDT

(B) $CHCl_3 + \frac{1}{2}O_2 \rightarrow$ (P) Phosgene gas

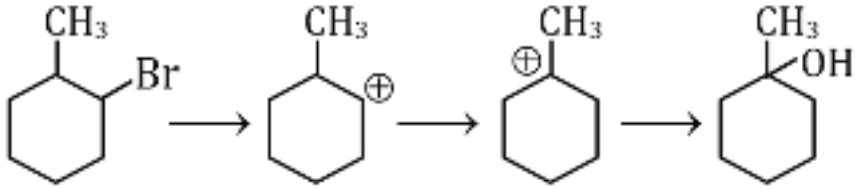
(C) $CH_4 + \underset{(excess)}{Cl_2} \xrightarrow{h\nu}$ (R) Fire exfinguisher

Final Answer : (2)

83) **Explaining :** Final product will be-

Concept : This question is based on reaction of Haloalkane with H_2O.

Solution :

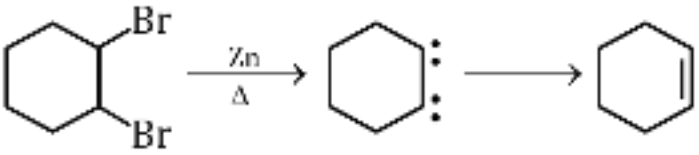


Final Answer : (3)

84) **Explaining :** Final product A is-

Concept : This question is based on dehalogenation.

Solution :



Final Answer : (3)

85) **Explaining :** Out of above reactions the total reactions which gives ether as a major product

Concept : This question is based on Ether preparation.

Solution :

E2 → a, b, c

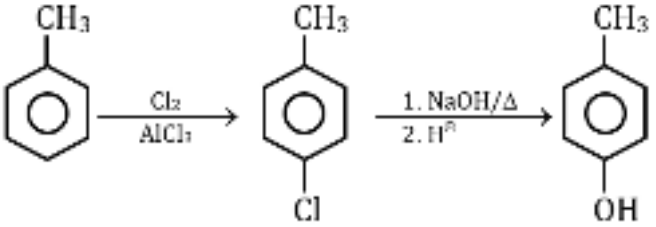
S_N2 → d, e

Final Answer : (2)

86) **Explaining :** Final product.

Concept : This question is based on Halogenation then SnAr reaction.

Solution :



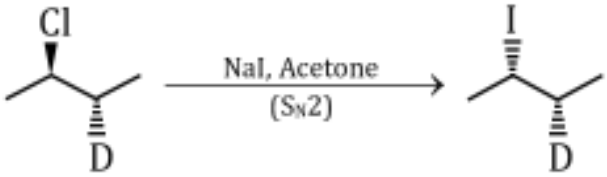
Final Answer : (2)

87) **Explaining :** Find product will be-

Concept : This question is based on S_N2 reaction.

Solution :

Inversion product



Final Answer : (4)

88) **Explaining :** The correct statement regarding the following two reactions.

(A) $CH_3 - CH_2 - Cl \xrightarrow{\bar{O}H}$

(B) $CH_3 - CH_2 - Cl \xrightarrow{\bar{S}H}$

Concept : This question is based on NSR mechanism.

Solution :

1° RX $\xrightarrow{\overset{\ominus}{O}H}$ S_N2 path

Nucleophilicity = $\overset{\ominus}{S}H > \overset{\ominus}{O}H$

Final Answer : (3)

89) **Explaining :** In which reaction halide is not obtained

(a) $CH_3-CH_2-CH_2-OH \xrightarrow{H-Cl}$

(b) $CH_3-CH_2-CH_2-OH \xrightarrow{H-Br}$

(c) $CH_3-CH_2-CH_2-OH \xrightarrow{H-I}$

(d) $Ph-OH \xrightarrow{H-Br}$

Concept : This question is based on prep. of R–X

Solution :

(a) 1° ROH $\xrightarrow{HCl}$ No reaction

(b) 1° ROH $\xrightarrow{HBr}$ Reaction possible

(c) 1° ROH $\xrightarrow{HI}$ Reaction possible

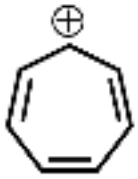
(d) Phenol $\xrightarrow{HBr}$ No reaction

Final Answer : (2)

90) **Explaining :** Which is maximum reactive towards H–Br

Concept : This question is based on stability of carbocation.

Solution :


Maxi. stable

Final Answer : (3)

BIOLOGY

91) **Solution :** Sperm formed in seminiferous tubules moves through:
Seminiferous tubules → rete testis → vasa efferentia → epididymis → vas deferens → ejaculatory duct → urethra.

Final Answer : (1)

92) **Solution : Sertoli cells** provide nutrition and support to developing germ cells. They are located inside seminiferous tubules.

Final Answer : (2)

93) **Solution:** If fertilisation does not occur, the corpus luteum degenerates. The resulting fall in progesterone causes degeneration of the endometrium, leading to menstruation.

Final Answer : (1)

94) **Solution:** Leydig cells, also called **interstitial cells**, occur in the interstitial spaces between seminiferous tubules and secrete **androgens, mainly testosterone**.

Final Answer : (2)

95) **Solution:** Primary spermatocytes are **diploid (2n)** and undergo **meiosis I** to form two

haploid secondary spermatocytes.
Final Answer : (4)

96) **Solution :**
Primary spermatocyte
↓ Meiosis I
2 secondary spermatocytes
↓ Meiosis II
4 spermatids
↓ Spermiogenesis
4 spermatozoa
Therefore, **one primary spermatocyte → four spermatozoa**.
Final Answer : (3)

97) **Solution:** The assertion is false. **LH surge**, not FSH, is primarily responsible for ovulation. The reason is true.
Final Answer : (4)

98) **Solution :** A-2, B-3, C-1, D-4

Structure	Function
A. Acrosome	2. Enzymes for sperm entry
B. Middle piece	3. Energy production
C. Tail	1. Sperm movement
D. Endometrium	4. Implantation

Final Answer : (2)

99) **Solution: Spermiogenesis** is the process by which **spermatids transform into spermatozoa**.
Final Answer : (2)

100) **Solution:** The **acrosome** develops from the **Golgi apparatus** and contains enzymes that help sperm penetrate the ovum.
Final Answer : (2)

101) **Solution:** The **middle piece** contains numerous mitochondria that provide energy for sperm movement.
Final Answer : (3)

102) **Solution: LH**, also called **ICSH in males**, acts on Leydig cells and stimulates androgen secretion.
FSH, in association with androgens, acts mainly on Sertoli cells and supports spermatogenesis.
Final Answer : (2)

103) **Solution:** The placenta performs exchange of **nutrients, gases and wastes** and also acts as an endocrine tissue. It has nothing to do with sperm production.
Final Answer : (4)

104) **Solution:** Fertilisation normally occurs at the **ampullary-isthmic junction** of the fallopian tube.
Final Answer : (3)

105) **Solution:** The **endometrium** undergoes cyclic changes and becomes thickened to prepare for implantation.
Final Answer : (3)

106) **Solution:** In a typical **28-day cycle**, ovulation occurs around **day 14** due to the **LH surge**.
Final Answer : (3)

107) **Solution:** A sudden increase in LH, called the **LH surge**, induces **ovulation**.
Final Answer : (2)

108) **Solution:** After ovulation, the corpus luteum secretes **progesterone**, which maintains the endometrium for possible implantation.
Final Answer : (3)

109) **Solution:** Progesterone maintains the uterine endometrium and supports implantation and early pregnancy. The corpus luteum is a major source of progesterone during the early phase.
Final Answer : (1)

110) **Solution :** Spermatogenesis begins at puberty — **True**
Spermatogonia are diploid — **True**
Secondary spermatocytes are haploid — so statement C is **False**
Spermatids are haploid — **True**
Final Answer : (1)

111) **Solution:** Statements A, B and C are correct.
The corpus luteum primarily secretes **progesterone**, not testosterone.
Final Answer : (4)

112) **Solution:** Zygote → Cleavage → Morula → Blastocyst → Implantation
The blastocyst attaches to the uterine endometrium during implantation.
Final Answer : (3)

113) **Solution:** Implantation occurs in the **uterine endometrium**, not the ovary.

Final Answer : (4)

114) **Solution :** Sertoli → nutrition/support
Leydig → testosterone
Corpus luteum → progesterone
Placenta → hCG
Final Answer : (4)

115) **Solution:** Follicle develops → estrogen rises → LH surge → ovulation → corpus luteum formation.
Final Answer : (1)

116) **Solution :** Spermatogonia → Primary spermatocyte → Secondary spermatocyte → Spermatid → Spermatozoa.
Final Answer : (1)

117) **Solution:** Without fertilisation, the corpus luteum degenerates → **progesterone decreases** → endometrium breaks down → menstruation occurs.
Final Answer: (2)

118) **Solution:** The placenta acts as an endocrine tissue and produces **hCG, hPL, estrogens and progesterone**.
Final Answer : (1)

119) **Solution:** The primary spermatocyte is **2n**. During **meiosis I**, it forms haploid secondary spermatocytes.
Therefore : **2n → n occurs during meiosis I.**
Final Answer : (2)

120) **Solution:** After fertilisation, the developing embryo produces **hCG**, which maintains the corpus luteum.
If implantation does not occur, adequate hCG-mediated maintenance of the corpus luteum does not occur. The corpus luteum degenerates, progesterone falls, and the endometrium is eventually shed.
Implantation → hCG → Corpus luteum maintained → Progesterone maintained → Endometrium maintained
Final Answer : (3)

121) **Solution:** In double-stranded B-DNA, the pitch of the helix is 3.4nm (34 AA) and there are roughly 10 base pairs per turn. Therefore, the distance between two consecutive base pairs is approximately 3.4nm (34 AA).
Final Answer : (2)

122) **Solution:** Statement 1 is correct as histones are basic proteins rich in lysine and arginine carrying positive charges. Statement 2 is incorrect because histones themselves are positively

charged, whereas the DNA molecule carries a negative charge and wraps around the positively charged histone octamer.
Final Answer : (3)

123) **Solution:** Hershey and Chase grew some viruses on a medium containing radioactive phosphorus (^{32}P) to label viral DNA (since phosphorus is present in DNA but not protein) and others on radioactive sulfur (^{35}S) to label viral proteins.
Final Answer : (3)

124)
Solution: RNA is chemically reactive and labile because the ribose sugar possesses a 2'-OH group at every nucleotide position. This makes RNA unstable and easily degradable, whereas DNA lacks this 2'-OH group, making it chemically more stable as a genetic material.
Final Answer : (1)

125) **Solution:** Francis Crick proposed the Central Dogma in molecular biology, which outlines the directional flow of genetic information: DNA replicates itself, undergoes transcription to form mRNA, and mRNA undergoes translation to synthesize proteins.
Final Answer : (2)

126) **Solution:** In eukaryotic nucleus, RNA polymerase I transcribes 28S, 18S and 5.85rRNAs; RNA polymerase II transcribes heterogeneous nuclear RNA (hnRNA); and RNA polymerase III transcribes tRNA, 5SrRNA, and small nuclear RNAs (snRNAs).
Final Answer : (1)

127) **Solution:** Capping is a modification where an unusual nucleotide, methyl guanosine triphosphate (7mGcap), is added to the *5*'-end of hnRNA. Tailing involves adding *200--300* adenylate residues at the *3*'-end.
Final Answer : (2)

128) **Solution:** Statements 1, 2, and 4 are correct. Statement 3 is incorrect because degeneracy means that **some amino acids are coded by more than one codon** (e.g., leucine, arginine), not that all amino acids are coded by a single codon.
Final Answer : (3)

129) **Solution:** The largest known human gene is dystrophin, which spans 2.4 million base pairs. On average, a human gene consists of 3000 bases, but sizes vary greatly.
Final Answer : (1)

130) **Solution:** In *E. coli*, the *lac* operon consists of one regulatory gene (gene) and three structural genes (z, y, a) that transcribe a polycistronic mRNA needed for lactose metabolism.
Final Answer : (4)

131) **Solution:** Francis Crick postulated the presence of an adapter molecule that would on one hand read the genetic code and on the other hand bind to specific amino acids. This

adapter role was subsequently proven to be fulfilled by tRNA (transfer RNA).
Final Answer : (2)

132) **Solution:** Meselson and Stahl (1958) grew *E. coli* in a medium containing heavy isotope $^{15}NH_4$ Cl. They demonstrated semi-conservative replication by separating DNA samples of different densities via equilibrium density gradient centrifugation using cesium chloride CsCl.
Final Answer : (1)

133) **Solution:** AUG serves as the primary start codon (initiator codon) in mRNA for translation and specifies the amino acid Methionine. UAA, UAG, and UGA are non-sense or stop codons that terminate translation.
Final Answer : (3)

134) **Solution:** According to HGP results, the human genome contains 3.1647 billion bp. Chromosome 1 has the highest number of genes (2968), whereas the Y chromosome has the fewest genes (231).
Final Answer : (4)

135) **Solution:** DNA fingerprinting utilizes Variable Number of Tandem Repeats (VNTRs), which belong to a class of satellite DNA called mini-satellites. The size of VNTR varies from individual to individual (0.1 to 20 kb), exhibiting a high degree of polymorphism used for forensic identification.
Final Answer : (2)

136) **Solution:** Frederick Griffith used *Streptococcus pneumoniae* in his transformation experiments. The S strain (smooth) produces a mucous polysaccharide coat and is virulent (causes fatal pneumonia in mice), whereas the R strain (rough) lacks the coat and is non-virulent.
Final Answer : (1)

137) **Solution:** DNA polymerase synthesizes DNA only in the 5' → 3' direction. On the template with 3' → 5' polarity, synthesis is continuous (leading strand). On the 5' → 3' template, synthesis is discontinuous (lagging strand), producing Okazaki fragments that are joined together by DNA ligase.
Final Answer : (2)

138) **Solution:** In bacteria, the 23S rRNA component of the large ribosomal subunit functions as a ribozyme (peptidyl transferase). It catalyzes the formation of peptide bonds between adjacent amino acids during the elongation phase of translation.
Final Answer : (4)

139) **Solution:** Amino acids are activated in the presence of ATP and linked to their specific tRNAs in a process called charging of tRNA. This supplies the activation energy needed for peptide bond formation when two charged tRNAs are positioned adjacent to each other in the ribosome.

Final Answer : (1)

140) **Solution:** Lactose (or allolactose) acts as the inducer in the *lac* operon. It binds to the lac repressor protein, altering its conformation so it cannot bind to the operator region, thereby allowing RNA polymerase to initiate transcription.
Final Answer : (3)

141) **Solution:** Statements 1, 2, and 4 are correct NCERT facts. Statement 3 is incorrect because Adenine forms **two** hydrogen bonds with Thymine (A = T), and Guanine forms **three** hydrogen bonds with Cytosine (G ≡ C).
Final Answer : (3)

142) **Solution:** The technique of DNA fingerprinting was developed by Sir Alec Jeffreys. He utilized satellite DNA as a radioactive probe showing high degree of polymorphism, termed Variable Number of Tandem Repeats (VNTRs).
Final Answer : (1)

143) **Solution:** Avery, MacLeod, and McCarty purified biochemicals to prove the transforming principle was DNA. Hershey and Chase provided unequivocal proof using bacteriophages. Jacob and Monod elucidated the *lac* operon. Nirenberg synthesized proteins in a cell-free system to crack the genetic code.
Final Answer : (1)

144) **Solution:** The strand with 3' → 5' polarity acts as the template strand during transcription. The other strand with 5' → 3' polarity does not code for anything, has the same sequence as the synthesized RNA (with T instead of U), and is designated as the coding strand.
Final Answer : (2)

145) **Solution:** All four statements are correct NCERT facts. HGP used two main approaches: ESTs (focused on expressed genes) and Sequence Annotation (sequencing the complete genome). Sanger's method was automated for large-scale sequencing, and fragments were cloned into BAC (Bacterial Artificial Chromosome) and YAC (Yeast Artificial Chromosome) vectors.
Final Answer : (4)

146) **Solution:** Mendel investigated 7 pairs of contrasting traits in garden pea plants (such as stem height, seed color, seed shape, pod color, pod shape, flower color, and flower position) involving 14 true-breeding pea plant varieties.
Final Answer : (1)

147) **Solution:** The Law of Segregation states that during gamete formation, the alleles of a gene segregate so that each gamete carries only one allele. Because blending does not occur, this law is considered universal without exceptions in diploid organisms.
Final Answer : (1)

148) **Solution:** This is a monohybrid test cross between a heterozygous tall (Tt) and a homozygous recessive dwarf (tt). The gametes produced are T and t from one parent and only t from the other, resulting in Tt (tall) and tt (dwarf) in a 1 : 1 ratio.
Final Answer : (2)

149) **Solution:** ABO blood groups are controlled by the gene I. I^A and I^B alleles produce slightly different sugar polymers on RBC membranes. When both I^A and I^B are present ($I^A I^B$), both express their respective sugars, demonstrating codominance.
Final Answer : (1)

150) **Solution:** Incomplete dominance in snapdragon results in pink flowers in the F_1 generation (Rr). Selfing F_1 plants yields F_2 offspring with Red (RR), Pink (Rr), and White (rr) in a 1 : 2 : 1 genotypic and phenotypic ratio.
Final Answer: (2)

151) **Solution:** Sickle-cell anemia is an autosomal recessive disease; Haemophilia is an X-linked sex-linked recessive disorder; Myotonic dystrophy is an autosomal dominant trait; and Klinefelter's syndrome is a chromosomal disorder caused by an extra X chromosome (47, XXY).
Final Answer : (4)

152) **Solution:** The test cross RrYy x rryy produces four genotypes: RrYy (round yellow), Rryy (round green), rrYy (wrinkled yellow), and rryy (wrinkled green) in a 1:1:1:1 ratio. The Parental types are 50% (RrYy and rryy) and Recombinant types are 50% (Rryy and rrYy).
Final Answer : (2)

153) **Solution :** All four statements are correct NCERT facts. Male heterogamety occurs in humans/Drosophila (□□) and grasshoppers (□□), while female heterogamety occurs in birds (□□). Honeybee males (drones) are haploid (□=16) and develop parthenogenetically.
Final Answer : (4)

154) **Solution :** Turner's syndrome is caused by the absence of one X chromosome in females (45,□□). Affected females are sterile with rudimentary ovaries, short stature, and lack of secondary sexual characters.
Final Answer : (2)

155) **Solution:** All four statements are direct facts from NCERT. T.H. Morgan mapped genes on the X chromosome of Drosophila. Sturtevant used Morgan's recombination frequencies to construct the first genetic map of chromosomes.
Final Answer : (4)

156) **Solution :** Sickle-cell anemia is caused by a point mutation resulting in the substitution of Glutamic acid (GAG) by Valine (GUG) at the sixth position of the □-globin chain of hemoglobin.
Final Answer : (1)

157) **Solution :** Phenylketonuria (PKU) is an autosomal recessive metabolic disorder caused by a mutation in the gene coding for the enzyme phenylalanine hydroxylase. Phenylalanine accumulates and converts into phenylpyruvic acid, causing mental retardation and excretion in urine.
Final Answer : (1)

158) **Solution :** The number of different types of gametes produced by a genotype is given by the formula 2□, where □ is the number of heterozygous gene pairs. Here, □ = 3 (□□,□□,□□), so 23 = 8 distinct gamete types are formed.
Final Answer : (2)

159) **Solution :** Sutton and Boveri proposed the Chromosomal Theory of Inheritance; Henking discovered the X-body (X chromosome) in insects; Human skin color is a classic example of polygenic inheritance; and Phenylketonuria exemplifies pleiotropy (single gene affecting multiple phenotypic traits).
Final Answer : (3)

160) **Solution :** Down's syndrome is caused by the presence of an extra copy of chromosome 21 (Trisomy 21), first described by Langdon Down in 1866. Symptoms include short stature, small round head, furrowed tongue, and mental development retardation.
Final Answer : (2)

161) **Solution :** Thalassemia is an autosomal recessive quantitative disorder of hemoglobin synthesis. α-Thalassemia is controlled by HBA1 and HBA2 genes on chromosome 16, whereas β-Thalassemia is controlled by a single HBB gene on chromosome 11.
Final Answer : (1)

162) **Solution :** Mother's genotype is ii (producing i gametes) and father's genotype is I^AI^B (producing I^A and I^B gametes). The offspring genotypes will be I^Ai (Blood Group A) and I^Bi (Blood Group B) in equal proportions.
Final Answer : (2)

163) **Solution:** All four statements are correct. Sickle-cell anemia is a qualitative defect (structural change in β-chain), unlike Thalassemia which is a quantitative defect (reduced synthesis of globin chains).
Final Answer : (4)

164) **Solution :** In the F_2 generation of a monohybrid cross (Tt x Tt), the genotypes are 1TT : 2Tt : 1tt. 25% are homozygous dominant (TT) and 25% are homozygous recessive (tt), making a total of 50% homozygous plants.
Final Answer : (2)

165) **Solution:** Drones develop from unfertilized haploid eggs via parthenogenesis, so they have no father. However, the queen that laid the egg developed from a fertilized egg (having a father, which is the drone's grandfather). Thus, males have no father/sons but have a

grandfather/grandsons.
Final Answer : (1)

166) **Solution:** In garden pea, Green pod color is dominant over Yellow pod color. Round seeds, axial flowers, and tall stem height are all dominant traits.
Final Answer : (1)

167) **Solution:** All four statements are standard NCERT conventions used in human pedigree analysis. A line between a square and circle represents mating, and double lines represent consanguineous mating (mating between relatives).
Final Answer : (4)

168) **Solution:** Aneuploidy is the condition of having an abnormal chromosome number due to non-disjunction of chromatids during cell division (e.g., Down's syndrome, Turner's syndrome). Failure of cytokinesis after telophase results in polyploidy.
Final Answer : (1)

169) **Solution:** Standard Mendelian ratios: Monohybrid phenotype = 3 : 1; Monohybrid genotype = 1 : 2 : 1; Dihybrid phenotype = 9 : 3 : 3 : 1; Dihybrid test cross = 1 : 1 : 1 : 1.
Final Answer : (2)

170) **Solution:** The mother produces two types of eggs: X^H (50%) and X^h (50%). The father passes his Y chromosome to all sons. Sons receiving the X^h chromosome (50% of total sons) will suffer from haemophilia (X^hY).
Final Answer : (3)

171) **Solution:** Starch synthesis in pea seeds exhibits dominance for seed shape (BB and Bb are round) but incomplete dominance for starch grain size (BB produces large grains, bb small grains, and Bb intermediate-sized grains).
Final Answer : (1)

172) **Solution:** From selfing Aa x Aa, probability of AA = 1/4.
From Bb x Bb, probability of bb = 1/4.
From Cc x Cc, probability of Cc = 1/4.
Combined probability for Aabbcc = (1/4) x (1/4) x (1/4) = 1/64.
Final Answer : (1)

173) **Explain Question –** The question asks us to match the labeled components (A–E) in the lac operon diagram with their correct identities: enzymes, repressor, and inducer.

Concept – Lac Operon – Regulation of Gene Expression in Prokaryotes

Solution –
Genes z, y, a code for enzymes:

z → A → β-galactosidase
y → B → Permease
a → C → Transacetylase
D → Repressor protein (coded by regulatory gene i)
E → Inducer (lactose) – binds to repressor to inactivate it

Final Answer – Option 1

174) **Solution :** Initiator codon is 5' AUG 3'
Final Answer : (2)

175) **Solution :** Introns are removed during splicing
Final Answer : (4)

176) **Solution :** Thymine is found in DNA only.
Final Answer : (3)

177) **Solution :** Histone octamer is made up of 8 histone units and linker protein joints two nucleosomes.
Final Answer : (1)

178) **Solution :** Male birds are homogametic.
Final Answer : (2)

179)

Explain Question: The question asks to analyse the image and answer the given options.
Concept: Dihybrid cross theory.
Solution:
The given figure explains:
I – Law of segregation
II – Law of Independent assortment

Final answer: 3.

180)

Question Explaination : The question asks to find the odd one wrt phenotype.

Concept : Dihybrid Cross.

Solution:

G : RRYy= Round Yellow

K: RRyy= Round Green

H: RrYY= Round Yellow

L: RrYy= Round Yellow

So, the correct answer is: 2. K

Final answer: 2.